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OPTICAL IMAGING SYSTEM

Abstract

An optical imaging system includes a first lens, a second lens, a third lens, a fourth lens, a fifth lens, a sixth lens, and a seventh lens sequentially disposed in numerical order along an optical axis of the optical imaging system from an object side of the optical imaging system toward an imaging plane of the optical imaging system, wherein the optical imaging system satisfies $0.5 < L_{12345TRavg} / L_{7TR} < 0.9$, where $L_{12345TRavg}$ is an average value of overall outer diameters of the first to fifth lenses, L_{7TR} is an overall outer diameter of the seventh lens, and $L_{12345TRavg}$ and L_{7TR} are expressed in a same unit of measurement.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/670,871 filed on May 22, 2024, which is a continuation of U.S. patent application Ser. No. 18/330,645 filed on Jun. 7, 2023, now U.S. Pat. No. 12,032,226, which is a continuation of U.S. patent application Ser. No. 17/088,665 filed on Nov. 4, 2020, now U.S. Pat. No. 11,714,263, which is a continuation of U.S. patent application Ser. No. 16/424,842 filed on May 29, 2019, now U.S. Pat. No. 11,016,271, which claims the benefit under 35 USC 119(a) of Korean Patent Application Nos. 10-2018-0061397 filed on May 29, 2018, and 10-2018-0106184 filed on Sep. 5, 2018, in the Korean Intellectual Property Office, the entire disclosures of which are incorporated herein by reference for all purposes.

BACKGROUND

1. Field

[0002] This application relates to an optical imaging system including seven lenses.

2. Description of Related Art

[0003] A mobile terminal is commonly provided with a camera for video communications or capturing images. However, it is difficult to achieve high performance in a camera for a mobile terminal due to space limitations inside the mobile terminal.

[0004] Accordingly, a demand for an optical imaging system capable of improving the performance of the camera without increasing a size of the camera has increased as a number of mobile terminals provided with a camera has increased.

SUMMARY

[0005] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

[0006] In one general aspect, an optical imaging system includes a first lens, a second lens, a third lens, a fourth lens, a fifth lens, a sixth lens, and a seventh lens sequentially disposed in numerical order along an optical axis of the optical imaging system from an object side of the optical imaging system toward an imaging plane of the optical imaging system, wherein the optical imaging system satisfies $0.5 < L_{12345TRavg}/L_{7TR} < 0.9$, where $L_{12345TRavg}$ is an average value of overall outer diameters of the first to fifth lenses, L_{7TR} is an overall outer diameter of the seventh lens, and $L_{12345TRavg}$ and L_{7TR} are expressed in a same unit of measurement.

[0007] An object-side surface of the first lens may be convex.

[0008] An image-side surface of the first lens may be concave.

[0009] An image-side surface of the seventh lens may be concave.

[0010] A distance along the optical axis from an object-side surface of the first lens to the imaging plane may be 6.0 mm or less.

[0011] At least one inflection point may be formed on either one or both of an object-side surface and an image-side surface of the sixth lens.

[0012] At least one inflection point may be formed on either one or both of an object-side surface and an image-side surface of the seventh lens.

[0013] The optical imaging system further may satisfy $0.1 < L1w/L7w < 0.3$, where $L1w$ is a weight of the first lens, $L7w$ is a weight of the seventh lens, and $L1w$ and $L7w$ are expressed in a same unit of measurement.

[0014] The optical imaging system may further include a spacer disposed between the sixth and seventh lenses, wherein the optical imaging system further may satisfy $0.5 < S6d/f < 1.2$, where $S6d$ is an inner diameter of the spacer, f is an overall focal length of the optical imaging system, and $S6d$ and f are expressed in a same unit of measurement.

[0015] The optical imaging system further may satisfy $0.4 < L1TR/L7TR < 0.7$, where $L1TR$ is an overall outer diameter of the first lens, and $L1TR$ and $L7TR$ are expressed in a same unit of measurement.

[0016] The optical imaging system further may satisfy $0.5 < L1234TRavg/L7TR < 0.75$, where $L1234TRavg$ is an average value of overall outer diameters of the first to fourth lenses, and $L1234TRavg$ and $L7TR$ are expressed in a same unit of measurement.

[0017] The optical imaging system further may satisfy $0.5 < L12345TRavg/L7TR < 0.76$.

[0018] The second lens may have a positive refractive power.

[0019] The third lens may have a positive refractive power.

[0020] The fifth lens may have a negative refractive power.

[0021] A paraxial region of an object-side surface of the fifth lens may be concave or convex.

[0022] A paraxial region of an image-side surface of the fifth lens may be concave or convex.

[0023] A paraxial region of an object-side surface of the sixth lens may be concave or convex.

[0024] A paraxial region of an image-side surface of the sixth lens may be concave or convex.

[0025] A paraxial region of an object-side surface of the seventh lens may be concave.

[0026] Other features and aspects will be apparent from the following detailed description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0027] FIG. 1 is a view illustrating a first example of an optical imaging system.

[0028] FIG. 2 illustrates aberration curves of the optical imaging system of FIG. 1.

[0029] FIG. 3 is a view illustrating a second example of an optical imaging system.

[0030] FIG. 4 illustrates aberration curves of the optical imaging system of FIG. 3.

[0031] FIG. 5 is a view illustrating a third example of an optical imaging system.

[0032] FIG. 6 illustrates aberration curves of the optical imaging system of FIG. 5.

[0033] FIG. 7 is a view illustrating a fourth example of an optical imaging system.

[0034] FIG. 8 illustrates aberration curves of the optical imaging system of FIG. 7.

[0035] FIG. 9 is a view illustrating a fifth example of an optical imaging system.

[0036] FIG. 10 illustrates aberration curves of the optical imaging system of FIG. 9.

[0037] FIG. 11 is a view illustrating a sixth example of an optical imaging system.

[0038] FIG. 12 illustrates aberration curves of the optical imaging system of FIG. 11.

[0039] FIG. 13 is a view illustrating a seventh example of an optical imaging system.

[0040] FIG. 14 illustrates aberration curves of the optical imaging system of FIG. 13.

[0041] FIG. 15 is a view illustrating an eighth example of an optical imaging system.

[0042] FIG. **16** illustrates aberration curves of the optical imaging system of FIG. **15**.
[0043] FIG. **17** is a view illustrating a ninth example of an optical imaging system.
[0044] FIG. **18** illustrates aberration curves of the optical imaging system of FIG. **17**.
[0045] FIG. **19** is a view illustrating a tenth example of an optical imaging system.
[0046] FIG. **20** illustrates aberration curves of the optical imaging system of FIG. **19**.
[0047] FIG. **21** is a view illustrating an eleventh example of an optical imaging system.
[0048] FIG. **22** illustrates aberration curves of the optical imaging system of FIG. **21**.
[0049] FIG. **23** is a view illustrating a twelfth example of an optical imaging system.
[0050] FIG. **24** illustrates aberration curves of the optical imaging system of FIG. **23**.
[0051] FIG. **25** is a view illustrating a thirteenth example of an optical imaging system.
[0052] FIG. **26** illustrates aberration curves of the optical imaging system of FIG. **25**.
[0053] FIG. **27** is a view illustrating a fourteenth example of an optical imaging system.
[0054] FIG. **28** illustrates aberration curves of the optical imaging system of FIG. **27**.
[0055] FIG. **29** is a view illustrating a fifteenth example of an optical imaging system.
[0056] FIG. **30** illustrates aberration curves of the optical imaging system of FIG. **29**.
[0057] FIG. **31** is a view illustrating a sixteenth example of an optical imaging system.
[0058] FIG. **32** illustrates aberration curves of the optical imaging system of FIG. **31**.
[0059] FIG. **33** is a view illustrating a seventeenth example of an optical imaging system.
[0060] FIG. **34** illustrates aberration curves of the optical imaging system of FIG. **33**.
[0061] FIG. **35** is a view illustrating an eighteenth example of an optical imaging system.
[0062] FIG. **36** illustrates aberration curves of the optical imaging system of FIG. **35**.
[0063] FIG. **37** is a view illustrating a nineteenth example of an optical imaging system.
[0064] FIG. **38** illustrates aberration curves of the optical imaging system of FIG. **37**.
[0065] FIG. **39** is a view illustrating a twentieth example of an optical imaging system.
[0066] FIG. **40** illustrates aberration curves of the optical imaging system of FIG. **39**.
[0067] FIG. **41** is a view illustrating a twenty-first example of an optical imaging system.
[0068] FIG. **42** illustrates aberration curves of the optical imaging system of FIG. **41**.
[0069] FIG. **43** is a view illustrating a twenty-second example of an optical imaging system.
[0070] FIG. **44** illustrates aberration curves of the optical imaging system of FIG. **43**.
[0071] FIG. **45** is a view illustrating a twenty-third example of an optical imaging system.
[0072] FIG. **46** illustrates aberration curves of the optical imaging system of FIG. **45**.
[0073] FIG. **47** is a view illustrating a twenty-fourth example of an optical imaging system.
[0074] FIG. **48** illustrates aberration curves of the optical imaging system of FIG. **47**.
[0075] FIG. **49** is a view illustrating a twenty-fifth example of an optical imaging system.
[0076] FIG. **50** illustrates aberration curves of the optical imaging system of FIG. **49**.
[0077] FIG. **51** is a view illustrating a twenty-sixth example of an optical imaging system.
[0078] FIG. **52** illustrates aberration curves of the optical imaging system of FIG. **51**.
[0079] FIG. **53** is a view illustrating a twenty-seventh example of an optical imaging system.
[0080] FIG. **54** illustrates aberration curves of the optical imaging system of FIG. **53**.
[0081] FIG. **55** is a view illustrating a twenty-eighth example of an optical imaging system.
[0082] FIG. **56** illustrates aberration curves of the optical imaging system of FIG. **55**.
[0083] FIG. **57** is a view illustrating a twenty-ninth example of an optical imaging system.
[0084] FIG. **58** illustrates aberration curves of the optical imaging system of FIG. **57**.
[0085] FIGS. **59** and **60** are cross-sectional views illustrating examples of an optical imaging system and a lens barrel coupled to each other.
[0086] FIG. **61** is cross-sectional view illustrating an example of a seventh lens.
[0087] FIG. **62** is a cross-sectional view illustrating an example of a shape of a rib of a lens.
[0088] Throughout the drawings and the detailed description, the same reference numerals refer to the same elements. The drawings may not be to scale, and the relative size, proportions, and depiction of elements in the drawings may be exaggerated for clarity, illustration, and convenience.

DETAILED DESCRIPTION

[0089] The following detailed description is provided to assist the reader in gaining a comprehensive understanding of the methods, apparatuses, and/or systems described herein. However, various changes, modifications, and equivalents of the methods, apparatuses, and/or systems described herein will be apparent after an understanding of the disclosure of this application. For example, the sequences of operations described herein are merely examples, and are not limited to those set forth herein, but may be changed as will be apparent after an understanding of the disclosure of this application, with the exception of operations necessarily occurring in a certain order. Also, descriptions of features that are known in the art may be omitted for increased clarity and conciseness.

[0090] The features described herein may be embodied in different forms, and are not to be construed as being limited to the examples described herein. Rather, the examples described herein have been provided merely to illustrate some of the many possible ways of implementing the methods, apparatuses, and/or systems described herein that will be apparent after an understanding of the disclosure of this application.

[0091] Throughout the specification, when an element, such as a layer, region, or substrate, is described as being “on,” “connected to,” or “coupled to” another element, it may be directly “on,” “connected to,” or “coupled to” the other element, or there may be one or more other elements intervening therebetween. In contrast, when an element is described as being “directly on,” “directly connected to,” or “directly coupled to” another element, there can be no other elements intervening therebetween.

[0092] As used herein, the term “and/or” includes any one and any combination of any two or more of the associated listed items.

[0093] Although terms such as “first,” “second,” and “third” may be used herein to describe various members, components, regions, layers, or sections, these members, components, regions, layers, or sections are not to be limited by these terms. Rather, these terms are only used to distinguish one member, component, region, layer, or section from another member, component, region, layer, or section. Thus, a first member, component, region, layer, or section referred to in examples described herein may also be referred to as a second member, component, region, layer, or section without departing from the teachings of the examples.

[0094] Spatially relative terms such as “above,” “upper,” “below,” and “lower” may be used herein for ease of description to describe one element's relationship to another element as shown in the figures. Such spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, an element described as being “above” or “upper” relative to another element will then be “below” or “lower” relative to the other element. Thus, the term “above” encompasses both the above and below orientations depending on the spatial orientation of the device. The device may also be oriented in other ways (for example, rotated by 90 degrees or at other orientations), and the spatially relative terms used herein are to be interpreted accordingly.

[0095] The terminology used herein is for describing various examples only, and is not to be used to limit the disclosure. The articles “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. The terms “comprises,” “includes,” and “has” specify the presence of stated features, numbers, operations, members, elements, and/or combinations thereof, but do not preclude the presence or addition of one or more other features, numbers, operations, members, elements, and/or combinations thereof.

[0096] Thicknesses, sizes, and shapes of lenses illustrated in the drawings may have been slightly exaggerated for convenience of explanation. In addition, the shapes of spherical surfaces or aspherical surfaces of the lenses described in the detailed description and illustrated in the drawings are merely examples. That is, the shapes of the spherical surfaces or the aspherical surfaces of the lenses are not limited to the examples described herein.

[0097] Numerical values of radii of curvature, thicknesses of lenses, distances between elements including lenses or surfaces, effective aperture radii of lenses, focal lengths, and diameters, thicknesses, and lengths of various elements are expressed in millimeters (mm), and angles are expressed in degrees. Thicknesses of lenses and distances between elements including lenses or surfaces are measured along the optical axis of the optical imaging system.

[0098] The term “effective aperture radius” as used in this application refers to a radius of a portion of a surface of a lens or other element (an object-side surface or an image-side surface of a lens or other element) through which light actually passes. The effective aperture radius is equal to a distance measured perpendicular to an optical axis of the surface between the optical axis of the surface and the outermost point on the surface through which light actually passes. Therefore, the effective aperture radius may be equal to a radius of an optical portion of a surface, or may be smaller than the radius of the optical portion of the surface if light does not pass through a peripheral portion of the optical portion of the surface. The object-side surface and the image-side surface of a lens or other element may have different effective aperture radii.

[0099] In this application, unless stated otherwise, a reference to the shape of a lens surface means the shape of a paraxial region of the lens surface. A paraxial region of a lens surface is a central portion of the lens surface surrounding the optical axis of the lens surface in which light rays incident to the lens surface make a small angle θ to the optical axis and the approximations $\sin \theta \approx \theta$, $\tan \theta \approx \theta$, and $\cos \theta \approx 1$ are valid.

[0100] For example, a statement that the object-side surface of a lens is convex means that at least a paraxial region of the object-side surface of the lens is convex, and a statement that the image-side surface of the lens is concave means that at least a paraxial region of the image-side surface of the lens is concave. Therefore, even though the object side-surface of the lens may be described as being convex, the entire object-side surface of the lens may not be convex, and a peripheral region of the object-side surface of the lens may be concave. Also, even though the image-side surface of the lens may be described as being concave, the entire image-side surface of the lens may not be concave, and a peripheral region of the image-side surface of the lens may be convex.

[0101] An optical imaging system includes a plurality of lenses. For example, the optical imaging system includes a first lens, a second lens, a third lens, a fourth lens, a fifth lens, a sixth lens, and a seventh lens sequentially disposed in numerical order along an optical axis of the optical imaging system from an object side of the optical imaging system toward an imaging plane of the optical imaging system. Thus, the first lens is a lens closest to an object (or a subject) to be imaged by the optical imaging system, while the seventh lens is a lens closest to the imaging plane.

[0102] Each lens of the optical imaging system includes an optical portion and a rib. The optical portion of the lens is a portion of the lens that is configured to refract light, and is generally formed in a central portion of the lens. The rib of the lens is an edge portion of the lens that enables the lens to be mounted in a lens barrel and the optical axis of the lens to be aligned with the optical axis of the optical imaging system. The rib of the lens extends radially outward from the optical portion. The optical portions of the lenses are generally not in contact with each other. For example, the first to seventh lenses are mounted in the lens barrel so that they are spaced apart from one another by predetermined distances along the optical axis of the optical imaging system. The ribs of the lenses may be in selective contact with each other. For example, the ribs of the first to fourth lenses, or the first to fifth lenses, or the second to fourth lenses, may be in contact with each other so that the optical axes of these lenses may be easily aligned with the optical axis of the optical imaging system.

[0103] Next, a configuration of the optical imaging system will be described.

[0104] The optical imaging system includes a plurality of lenses. For example, the optical imaging system includes a first lens, a second lens, a third lens, a fourth lens, a fifth lens, a sixth lens, and a seventh lens sequentially disposed in numerical order along an optical axis of the optical imaging system from an object side of the optical imaging system toward an imaging plane of the optical

imaging system.

[0105] The optical imaging system further includes an image sensor and a filter. The image sensor forms an imaging plane, and converts light refracted by the first to seventh lenses into an electric signal. The filter is disposed between the seventh lens and the imaging plane, and blocks infrared rays in the light refracted by the first to seventh lenses from being incident on the imaging plane.

[0106] The optical imaging system further includes a stop and spacers. The stop is disposed in front of the first lens, or at a position of either an object-side surface or an image side-surface of one of the first to seventh lenses, or between two adjacent lenses of the first to seventh lenses, or between the object-side surface and the image-side surface of one of the first to seventh lenses, to adjust the amount of light incident on the imaging plane. Some examples may include two stops, one of which may be disposed in front of the first lens, or at the position of the object-side surface of the first lens, or between the object-side surface and the image-side surface of the first lens. Each of the spacers is disposed at a respective position between two lenses of the first to seventh lenses to maintain a predetermined distance between the two lenses. In addition, the spacers may be made of a light-shielding material to block extraneous light penetrating into the ribs of the lenses. There may be six or seven spacers. For example, a first spacer is disposed between the first lens and the second lens, a second spacer is disposed between the second lens and the third lens, a third spacer is disposed between the third lens and the fourth lens, a fourth spacer is disposed between the fourth lens and the fifth lens, a fifth spacer is disposed between the fifth lens and the sixth lens, and a sixth spacer is disposed between the sixth lens and the seventh lens. In addition, the optical imaging system may further include a seventh spacer disposed between the sixth lens and the sixth spacer.

[0107] Next, the lenses of the optical imaging system will be described.

[0108] The first lens has a refractive power. For example, the first lens has a positive refractive power or a negative refractive power. One surface of the first lens may be convex. For example, an object-side surface of the first lens may be convex. One surface of the first lens may be concave. For example, an image-side surface of the first lens may be concave. The first lens may have an aspherical surface. For example, one surface or both surfaces of the first lens may be aspherical.

[0109] The second lens has a refractive power. For example, the second lens has a positive refractive power or a negative refractive power. At least one surface of the second lens may be convex. For example, an object-side surface of the second lens may be convex, or both the object-side surface and an image-side surface of the second lens may be convex. One surface of the second lens may be concave. For example, the image-side surface of the second lens may be concave. The second lens may have an aspherical surface. For example, one surface or both surfaces of the second lens may be aspherical.

[0110] The third lens has a refractive power. For example, the third lens has a positive refractive power or a negative refractive power. One surface of the third lens may be convex. For example, an object-side surface or an image-side surface of the third lens may be convex. One surface of the third lens may be concave. For example, the object-side surface or the image-side surface of the third lens may be concave. The third lens may have an aspherical surface. For example, one surface or both surfaces of the third lens may be aspherical.

[0111] The fourth lens has a refractive power. For example, the fourth lens has a positive refractive power or a negative refractive power. At least one surface of the fourth lens may be convex. For example, an object-side surface or an image-side surface of the fourth lens may be convex, or both the object-side surface and the image-side surface of the fourth lens may be convex. One surface of the fourth lens may be concave. For example, the object-side surface or the image-side surface of the fourth lens may be concave. The fourth lens may have an aspherical surface. For example, one surface or both surfaces of the fourth lens may be aspherical.

[0112] The fifth lens has a refractive power. For example, the fifth lens has a positive refractive power or a negative refractive power. One surface of the fifth lens may be convex. For example, an

object-side surface or an image-side surface of the fifth lens may be convex. At least one surface of the fifth lens may be concave. For example, the object-side surface or the image-side surface of the fifth lens may be concave, or both the object-side surface and the image-side surface of the fifth lens may be concave. The fifth lens may have an aspherical surface. For example, one surface or both surfaces of the fifth lens may be aspherical.

[0113] The sixth lens has a refractive power. For example, the sixth lens has a positive refractive power or a negative refractive power. At least one surface of the sixth lens may be convex. For example, an object-side surface or an image-side surface of the sixth lens may be convex, or both the object-side surface and the image-side surface of the sixth lens may be convex. At least one surface of the sixth lens may be concave. For example, the object-side surface or the image-side surface of the sixth lens may be concave, or both the object-side surface and the image-side surface of the sixth lens may be concave. At least one surface of the sixth lens may have at least one inflection point. An inflection point is a point where a lens surface changes from convex to concave, or from concave to convex. A number of inflection points is counted from a center of the lens to an outer edge of the optical portion of the lens. For example, at least one inflection point may be formed on either one or both of the object-side surface and the image-side surface of the sixth lens. Therefore, at least one surface of the sixth lens may have a paraxial region and a peripheral region having shapes that are different from each other. For example, a paraxial region of the object-side surface of the sixth lens may be concave, but a peripheral region thereof may be convex. The sixth lens may have an aspherical surface. For example, one surface or both surfaces of the sixth lens may be aspherical.

[0114] The seventh lens has a refractive power. For example, the seventh lens has a positive refractive power or a negative refractive power. One surface of the seventh lens may be convex. For example, an object-side surface of the seventh lens may be convex. At least one surface of the seventh lens may be concave. For example, an image-side surface of the seventh lens may be concave, or both the object-side surface and the image-side surface of the seventh lens may be concave. At least one surface of the seventh lens may have at least one inflection point. For example, at least one inflection point may be formed on either one or both of the object-side surface and the image-side surface of the seventh lens. Therefore, at least one surface of the seventh lens may have a paraxial region and a peripheral region having shapes that are different from each other. For example, a paraxial region of the object-side surface of the seventh lens may be concave, but a peripheral region thereof may be convex. The seventh lens may have an aspherical surface. For example, one surface or both surfaces of the seventh lens may be aspherical.

[0115] The lenses of the optical imaging system may be made of a light material having a high light transmittance. For example, the first to seventh lenses may be made of a plastic material. However, a material of the first to seventh lenses is not limited to the plastic material.

[0116] The aspherical surfaces of the first to seventh lenses may be represented by the following Equation 1:

[00001]

$$Z = \frac{cY^2}{1 + \sqrt{1 - (1 + K)c^2Y^2}} + AY^4 + BY^6 + CY^8 + DY^{10} + EY^{12} + FY^{14} + GY^{16} + HY^{18} + \sim \quad (1)$$

[0117] In Equation 1, c is a curvature of a lens surface and is equal to an inverse of a radius of curvature of the lens surface at an optical axis of the lens surface, K is a conic constant, Y is a distance from a certain point on an aspherical surface of the lens to an optical axis of the lens in a direction perpendicular to the optical axis, A to H are aspherical constants, and Z (or sag) is a distance between the certain point on the aspherical surface of the lens at the distance Y to the optical axis and a tangential plane perpendicular to the optical axis meeting the apex of the aspherical surface of the lens. Some of the examples disclosed in this application include an aspherical constant J. An additional term of JY^{sup.20} may be added to the right side of Equation 1 to reflect the effect of the aspherical constant J.

[0118] The optical imaging system may satisfy one or more of the following Conditional Expressions 1 to 5:

[00002] $0.1 < L1w / L7w < 0.4$ (ConditionalExpression1)

$0.5 < S6d / f < 1.4$ (ConditionalExpression2)

$0.4 < L1TR / L7TR < 0.8$ (ConditionalExpression3)

$0.5 < L1234TRavg / L7TR < 0.9$ (ConditionalExpression4)

$0.5 < L12345TRavg / L7TR < 0.9$ (ConditionalExpression5)

[0119] In the above Conditional Expressions, L1w is a weight of the first lens, and L7w is a weight of the seventh lens.

[0120] S6d is an inner diameter of the sixth spacer, and f is an overall focal length of the optical imaging system.

[0121] L1TR is an overall outer diameter of the first lens, and L7TR is an overall outer diameter of the seventh lens. The overall outer diameter of a lens is a diameter of the lens including both the optical portion of the lens and the rib of the lens.

[0122] L1234TRavg is an average value of overall outer diameters of the first to fourth lenses, and L12345TRavg is an average value of overall outer diameters of the first to fifth lenses.

[0123] Conditional Expressions 1 and 3 specify ranges of a weight ratio and an overall outer diameter ratio between the first lens and the seventh lens to facilitate a self-alignment between the lenses and an alignment by the lens barrel.

[0124] Conditional Expression 2 specifies a range of a ratio of the inner diameter of the sixth spacer to the overall focal length of the optical imaging system to minimize a flare phenomenon.

[0125] Conditional Expressions 4 and 5 specify overall outer diameter ratios between the lenses to facilitate aberration correction.

[0126] The optical imaging system may also satisfy one or more of the following Conditional Expressions 6 to 10:

[00003] $0.1 < L1w / L7w < 0.3$ (ConditionalExpression6)

$0.5 < S6d / f < 1.2$ (ConditionalExpression7)

$0.4 < L1TR / L7TR < 0.7$ (ConditionalExpression8)

$0.5 < L1234TRavg / L7TR < 0.75$ (ConditionalExpression9)

$0.5 < L12345TRavg / L7TR < 0.76$ (ConditionalExpression10)

[0127] Conditional Expressions 6 to 10 are the same as Conditional Expressions 1 to 5, except that Conditional Expressions 6 to 10 specify narrower ranges.

[0128] The optical imaging system may also satisfy one or more of the following Conditional Expressions 11 to 31:

[00004] $0.01 < R1 / R4 < 1.3$ (ConditionalExpression11)

$0.1 < R1 / R5 < 0.7$ (ConditionalExpression12)

$0.05 < R1 / R6 < 0.9$ (ConditionalExpression13)

$0.2 < R1 / R11 < 1.2$ (ConditionalExpression14)

$0.8 < R1 / R14 < 1.2$ (ConditionalExpression15)

$0.6 < (R11 + R14) / (2 * R1) < 3$. (ConditionalExpression16)

$0.4 < D13 / D57 < 1.2$ (ConditionalExpression17)

$0.1 < (1 / f1 + 1 / f2 + 1 / f3 + 1 / f4 + 1 / f5 + 1 / f6 + 1 / f7) * f < 0.8$ (ConditionalExpression18)

$0.1 < (1 / f1 + 1 / f2 + 1 / f3 + 1 / f4 + 1 / f5 + 1 / f6 + 1 / f7) * TTL < 1$. (ConditionalExpression19)

$0.2 < TD1 / D67 < 0.8$ (ConditionalExpression20)

$$0.1 < (R11 + R14) / (R5 + R6) < 1. \quad (\text{ConditionalExpression21})$$

$$SD12 < SD34 \quad (\text{ConditionalExpression22}) \quad SD56 < SD67 \quad (\text{ConditionalExpression23})$$

$$SD56 < SD34 \quad (\text{ConditionalExpression24})$$

$$0.6 < TTL / (2^* (IMGHT)) < 0.9 \quad (\text{ConditionalExpression25})$$

$$0.2 < SD / TD < 0.7 \quad (\text{ConditionalExpression26})$$

$$0 < \min(f1:f3) / \max(f4:f7) < 0.4 \quad (\text{ConditionalExpression27})$$

$$0.4 < (TD) / TTL < 0.7 \quad (\text{ConditionalExpression28})$$

$$0.7 < SL / TTL < 1. \quad (\text{ConditionalExpression29})$$

$$0.81 < f12 / f123 < 0.96 \quad (\text{ConditionalExpression30})$$

$$0.6 < f12 / f1234 < 0.84 \quad (\text{ConditionalExpression31})$$

[0129] In the above Conditional Expressions, R1 is a radius of curvature of an object-side surface of the first lens, R4 is a radius of curvature of an image-side surface of the second lens, R5 is a radius of curvature of an object-side surface of the third lens, R6 is a radius of curvature of an image-side surface of the third lens, R11 is a radius of curvature of an object-side surface of the sixth lens, and R14 is a radius of curvature of an image-side surface of the seventh lens.

[0130] D13 is a distance along an optical axis of the optical imaging system from the object-side surface of the first lens to the image-side surface of the third lens, and D57 is a distance along the optical axis from an object-side surface of the fifth lens to the image-side surface of the seventh lens.

[0131] f1 is a focal length of the first lens, f2 is a focal length of the second lens, f3 is a focal length of the third lens, f4 is a focal length of the fourth lens, f5 is a focal length of the fifth lens, f6 is a focal length of the sixth lens, f7 is a focal length of the seventh lens, f is an overall focal length of the optical imaging system, and TTL is a distance along the optical axis from the object-side surface of the first lens to an imaging plane of an image sensor of the optical imaging system.

[0132] TD1 is a thickness along the optical axis of the first lens, and D67 is a distance along the optical axis from the object-side surface of the sixth lens to the image-side surface of the seventh lens.

[0133] SD12 is a distance along the optical axis from an image-side surface of the first lens to an object-side surface of the second lens, SD34 is a distance along the optical axis from the image-side surface of the third lens to an object-side surface of the fourth lens, SD56 is a distance along the optical axis from an image-side surface of the fifth lens to the object-side surface of the sixth lens, and SD67 is a distance along the optical axis from an image-side surface of the sixth lens to an object-side surface of the seventh lens.

[0134] IMG HT is one-half of a diagonal length of the imaging plane of the image sensor.

[0135] Σ SD is a sum of air gaps along the optical axis between the lenses, and Σ TD is a sum of thicknesses along the optical axis of the lenses. An air gap is a distance along the optical axis between adjacent lenses.

[0136] $\min(f1:f3)$ is a minimum value of absolute values of the focal lengths of the first to third lenses, and $\max(f4:f7)$ is a maximum value of absolute values of the focal lengths of the fourth to seventh lenses.

[0137] SL is a distance along the optical axis from the stop to the imaging plane of the image sensor.

[0138] f12 is a composite focal length of the first and second lenses, f123 is a composite focal length of the first to third lenses, and f1234 is a composite focal length of the first to fourth lenses.

[0139] Conditional Expression 11 specifies a design range of the second lens for minimizing aberration caused by the first lens. For example, it is difficult to expect a sufficient correction of longitudinal spherical aberration for the second lens having a radius of curvature that exceeds the

upper limit value of Conditional Expression 11, and it is difficult to expect a sufficient correction of astigmatic field curves for the second lens having a radius of curvature that is below the lower limit value of Conditional Expression 11.

[0140] Conditional Expressions 12 and 13 specify a design range of the third lens for minimizing aberration caused by the first lens. For example, it is difficult to expect a sufficient correction of longitudinal spherical aberration for the third lens having a radius of curvature that exceeds the upper limit value of Conditional Expression 12 or 13, and it is difficult to expect a sufficient correction of astigmatic field curves for the third lens having a radius of curvature that is below the lower limit value of Conditional Expression 12 or 13.

[0141] Conditional Expression 14 specifies a design range of the sixth lens for minimizing aberration caused by the first lens. For example, it is difficult to expect a sufficient correction of longitudinal spherical aberration for the sixth lens having a radius of curvature that exceeds the upper limit value of Conditional Expression 14, and the sixth lens having a radius of curvature that is below the lower limit value of Conditional Expression 14 is apt to cause a flare phenomenon.

[0142] Conditional Expression 15 specifies a design range of the seventh lens for minimizing the aberration caused by the first lens. For example, it is difficult to expect a sufficient correction of longitudinal spherical aberration for the seventh lens having a radius of curvature that exceeds the upper limit value of Conditional Expression 15, and the seventh lens having a radius of curvature that is below the lower limit value of Conditional Expression 15 is apt to cause an imaging plane curvature.

[0143] Conditional Expression 16 specifies a ratio of a sum of radii of curvature of the sixth lens and the seventh lens to twice a radius of curvature of the first lens for correcting the longitudinal spherical aberration and achieving excellent optical performance.

[0144] Conditional Expression 17 specifies a ratio of the optical imaging system mountable in a compact terminal. For example, an optical imaging system having a ratio that exceeds the upper limit value of Conditional Expression 17 may cause a problem that the total length of the optical imaging system becomes long, and an optical imaging system having a ratio that is below the lower limit value of Conditional Expression 17 may cause a problem that a lateral cross-section of the optical imaging system becomes large.

[0145] Conditional Expressions 18 and 19 specify a refractive power ratio of the first to seventh lenses for facilitating mass production of the optical imaging system. For example, an optical imaging system having a refractive power ratio that exceeds the upper limit value of Conditional Expression 18 or 19 or is below the lower limit value of Conditional Expression 18 or 19 is difficult to commercialize because the refractive power of one or more of the first to seventh lenses is too great.

[0146] Conditional Expression 20 specifies a thickness range of the first lens for implementing a compact optical imaging system. For example, the first lens having a thickness that exceeds the upper value of Conditional Expression 20 or is below the lower limit value of Conditional Expression 20 is too thick or too thin to be manufactured.

[0147] Conditional Expression 22 specifies a design condition of the first to fourth lenses for improving chromatic aberration. For example, a case in which a distance between the first lens and the second lens is shorter than a distance between the third lens and the fourth lens is advantageous for improving the chromatic aberration.

[0148] Conditional Expressions 25 to 28 specify design conditions for implementing a compact optical imaging system. For example, lenses that deviate from the numerical range of Conditional Expression 26 or 28 are difficult to form by injection molding and process.

[0149] Conditional Expressions 29 to 31 specify design conditions of the optical imaging system in consideration of a position of the stop. For example, an optical imaging system that does not satisfy one or more of Conditional Expressions 29 to 31 may have a longer overall length due to the refractive power of the lenses disposed behind the stop.

Infinity 2.915 S17 Imaging Infinity 0.0120 3.268 Plane
TABLE-US-00002 TABLE 2 K A B C D E F G H S1 -0.0435 -0.0109 0.0629 -0.3496 0.8258
-1.1525 0.7549 -0.2137 0 S2 -13.409 -0.1278 0.1627 0.0433 -0.8903 1.7755 -1.5568 0.5056 0
S3 -3.4003 -0.1481 0.3107 -0.2639 0.1553 -0.0446 0.299 -0.2867 0 S4 0.21 -0.0544 0.1307
-0.1443 0.9025 -2.9213 4.8603 -2.7957 0 S5 3.2497 -0.0995 0.0696 -0.678 2.7904 -5.9532
7.488 -3.502 0 S6 0.0843 -0.0685 -0.0666 0.3302 -1.5683 4.6995 -6.2319 3.6509 0 S7 4E-08
-0.155 -0.0515 -0.2096 0.1445 0.5442 -0.6037 0.1711 0 S8 5E-09 -0.1321 0.0922 -0.6514
0.9947 -0.5706 0.1385 -0.0115 0 S9 -0.3203 -0.1327 0.4222 -1.1911 1.2725 -0.4351 -0.1439
0.0842 0 S10 10.535 -0.1849 0.2573 -0.4641 0.511 -0.2753 0.069 -0.0065 0 S11 -50 0.1562
-0.6443 0.8704 -0.7495 0.3744 -0.0961 0.0098 0 S12 -34.584 0.0511 -0.2768 0.3083 -0.2075
0.0819 -0.0171 0.0015 0 S13 -0.9404 -0.5384 0.2288 -0.0102 -0.0258 0.0106 -0.002 0.0002
-7E-06 S14 -0.996 -0.4669 0.3054 -0.1607 0.0625 -0.0162 0.0026 -0.0002 8E-06

Second Example

[0156] FIG. 3 is a view illustrating a second example of an optical imaging system, and FIG. 4 illustrates aberration curves of the optical imaging system of FIG. 3.

[0157] An optical imaging system 2 includes a first lens 1002, a second lens 2002, a third lens 3002, a fourth lens 4002, a fifth lens 5002, a sixth lens 6002, and a seventh lens 7002.

[0158] The first lens 1002 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens 2002 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. The third lens 3002 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens 4002 has a positive refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The fifth lens 5002 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens 6002 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens 6002. The seventh lens 7002 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens 7002, and one inflection point is formed on the image-side surface of the seventh lens 7002.

[0159] The optical imaging system 2 further includes a stop, a filter 8002, and an image sensor 9002. The stop is disposed between the second lens 2002 and the third lens 3002 to adjust an amount of light incident on the image sensor 9002. The filter 8002 is disposed between the seventh lens 7002 and the image sensor 9002 to block infrared rays. The image sensor 9002 forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 3, the stop is disposed at a distance of 1.425 mm from the object-side surface of the first lens 1002 toward the imaging plane of the optical imaging system 2. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 2 listed in Table 59 that appears later in this application.

[0160] Table 3 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 3, and Table 4 below shows aspherical surface coefficients of the lenses of FIG. 3.

TABLE-US-00003 TABLE 3 Effective Surface Radius of Thickness/ Index of Abbe Aperture No.
Element Curvature Distance Refraction Number Radius S1 First 2.3706 0.5431 1.546 56.114
1.572 S2 Lens 3.8377 0.1516 1.517 S3 Second 3.4329 0.7078 1.546 56.114 1.478 S4 Lens
-17.0251 0.0225 1.428 S5 Third Lens 5.1429 0.2247 1.679 19.236 1.300 (Stop) S6 2.5333
0.5888 1.230 S7 Fourth -1446.167 0.3404 1.679 19.236 1.404 S8 Lens -1446.167 0.2070 1.600
S9 Fifth 3.6434 0.3264 1.546 56.114 1.857 S10 Lens 3.8224 0.3171 2.199 S11 Sixth 3.8509
0.4406 1.679 19.236 2.415 S12 Lens 3.0494 0.1774 2.808 S13 Seventh 1.7430 0.6133 1.537

53.195 3.115 S14 Lens 1.5635 0.2466 3.314 S15 Filter Infinity 0.1100 1.519 64.166 3.655 S16
Infinity 0.8047 3.688 S17 Imaging Infinity 0.0051 4.075 Plane
TABLE-US-00004 TABLE 4 K A B C D E F G H J S1 -7.5196 0.0476 -0.039 0.0108 -0.0002
-0.006 0.0045 -0.0012 0.0001 0 S2 -19.661 -0.0106 -0.0481 0.0183 0.0105 -0.0109 0.0039
-0.0006 3E-05 0 S3 0.042 -0.0249 -0.0196 0.0094 0.0041 0.0108 -0.014 0.0056 -0.0008 0
S4 0 0.0098 -0.0507 0.0341 0.0229 -0.0518 0.0341 -0.0103 0.0012 0 S5 -5.6502 -0.0476
0.0152 -0.0398 0.11 -0.1327 0.082 -0.0252 0.0031 0 S6 0.5327 -0.067 0.0583 -0.0705 0.0922
-0.0854 0.0499 -0.0161 0.0024 0 S7 0 -0.0158 -0.0083 -0.0305 0.0756 -0.0736 0.035 -0.0077
0.0005 0 S8 0 -0.0099 -0.0427 0.0077 0.0285 -0.0272 0.01 -0.0013 0 0 S9 -44.395 0.1048
-0.1251 0.08 -0.0437 0.0187 -0.0058 0.001 -7E-05 0 S10 -4.0715 -0.0175 0.0211 -0.0368
0.0252 -0.01 0.0024 -0.0003 2E-05 0 S11 -1.1211 0.0034 -0.0742 0.0637 -0.0381 0.0134
-0.0026 0.0003 -1E-05 0 S12 0.0464 -0.092 0.0339 -0.0168 0.0044 -0.0005 1E-05 3E-06
-2E-07 0 S13 -0.795 -0.2987 0.11 -0.0259 0.0046 -0.0007 7E-05 -5E-06 2E-07 -5E-09 S14
-1.3233 -0.199 0.0846 -0.0285 0.0073 -0.0013 0.0002 -1E-05 5E-07 -9E-09

Third Example

[0161] FIG. 5 is a view illustrating a third example of an optical imaging system, and FIG. 6 illustrates aberration curves of the optical imaging system of FIG. 5.

[0162] An optical imaging system 3 includes a first lens 1003, a second lens 2003, a third lens 3003, a fourth lens 4003, a fifth lens 5003, a sixth lens 6003, and a seventh lens 7003.

[0163] The first lens 1003 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens 2003 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. The third lens 3003 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens 4003 has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The fifth lens 5003 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens 6003 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens 6003. The seventh lens 7003 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens 7003, and one inflection point is formed on the image-side surface of the seventh lens 7003.

[0164] The optical imaging system 3 further includes a stop, a filter 8003, and an image sensor 9003. The stop is disposed between the second lens 2003 and the third lens 3003 to adjust an amount of light incident on the image sensor 9003. The filter 8003 is disposed between the seventh lens 7003 and the image sensor 9003 to block infrared rays. The image sensor 9003 forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 5, the stop is disposed at a distance of 1.259 mm from the object-side surface of the first lens 1003 toward the imaging plane of the optical imaging system 3. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 3 listed in Table 59 that appears later in this application.

[0165] Table 5 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 5, and Table 6 below shows aspherical surface coefficients of the lenses of FIG. 5.

TABLE-US-00005 TABLE 5 Effective Surface Radius of Thickness/ Index of Abbe Aperture No.
Element Curvature Distance Refraction Number Radius S1 First 2.1022 0.4835 1.546 56.114
1.408 S2 Lens 3.3563 0.1357 1.350 S3 Second 3.0907 0.6198 1.546 56.114 1.308 S4 Lens
-13.9876 0.0200 1.271 S5 Third Lens 4.8553 0.2000 1.679 19.236 1.157 (Stop) S6 2.3669
0.5599 1.095 S7 Fourth -2272.129 0.3012 1.679 19.236 1.270 S8 Lens -7278.426 0.1848 1.442

S9 Fifth 3.3546 0.2946 1.546 56.114 1.646 S10 Lens 3.5201 0.2604 1.947 S11 Sixth 3.4723 0.3932 1.679 19.236 2.150 S12 Lens 2.7354 0.1549 2.500 S13 Seventh 1.5570 0.5518 1.537 53.955 2.749 S14 Lens 1.3661 0.2501 2.950 S15 Filter Infinity 0.1100 1.519 64.166 3.293 S16 Infinity 0.6646 3.328 S17 Imaging Infinity 0.0054 3.699 Plane

TABLE-US-00006 TABLE 6 K A B C D E F G H J S1 -7.5279 0.0685 -0.0723 0.0313 -0.0131 -0.0097 0.0144 -0.0054 0.0007 0 S2 -19.893 -0.0114 -0.0921 0.0405 0.0318 -0.0345 0.0116 -0.001 -0.0002 0 S3 -0.0142 -0.0359 -0.0288 -0.0087 0.0581 0.0053 -0.0505 0.0291 -0.0054 0 S4 0 0.0225 -0.1301 0.1638 -0.0413 -0.1012 0.1103 -0.0452 0.0067 0 S5 -6.2325 -0.061 -0.0037 -0.0472 0.3094 -0.5229 0.4199 -0.1649 0.0257 0 S6 0.4782 -0.092 0.0962 -0.1588 0.2881 -0.3518 0.2616 -0.1062 0.0192 0 S7 0 -0.0151 -0.0532 0.0425 0.0094 -0.0356 0.0085 0.009 -0.0039 0 S8 0 -0.0101 -0.0934 0.0497 0.0399 -0.0661 0.0321 -0.0053 0 0 S9 -49.08 0.1451 -0.2207 0.1683 -0.1105 0.058 -0.0226 0.0051 -0.0005 0 S10 -5.4303 -0.0164 0.0172 -0.0595 0.0534 -0.0275 0.0084 -0.0014 1E-04 0 S11 -1.136 0.0251 -0.1801 0.1935 -0.1377 0.0586 -0.014 0.0017 -9E-05 0 S12 0.0272 -0.1034 0.0166 3E-05 -0.0063 0.0037 -0.0009 0.0001 -5E-06 0 S13 -0.8 -0.4195 0.2062 -0.0728 0.0211 -0.0048 0.0007 -8E-05 4E-06 -1E-07 S14 -1.3207 -0.2931 0.1671 -0.0741 0.0239 -0.0053 0.0008 -7E-05 4E-06 -8E-08

Fourth Example

[0166] FIG. 7 is a view illustrating a fourth example of an optical imaging system, and FIG. 8 illustrates aberration curves of the optical imaging system of FIG. 7.

[0167] An optical imaging system 4 includes a first lens 1004, a second lens 2004, a third lens 3004, a fourth lens 4004, a fifth lens 5004, a sixth lens 6004, and a seventh lens 7004.

[0168] The first lens 1004 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens 2004 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. The third lens 3004 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens 4004 has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The fifth lens 5004 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens 6004 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens 6004. The seventh lens 7004 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens 7004, and one inflection point is formed on the image-side surface of the seventh lens 7004.

[0169] The optical imaging system 4 further includes a stop, a filter 8004, and an image sensor 9004. The stop is disposed between the second lens 2004 and the third lens 3004 to adjust an amount of light incident on the image sensor 9004. The filter 8004 is disposed between the seventh lens 7004 and the image sensor 9004 to block infrared rays. The image sensor 9004 forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 7, the stop is disposed at a distance of 1.169 mm from the object-side surface of the first lens 1004 toward the imaging plane of the optical imaging system 4. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 4 listed in Table 59 that appears later in this application.

[0170] Table 7 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 7, and Table 8 below shows aspherical surface coefficients of the lenses of FIG. 7.

TABLE-US-00007 TABLE 7 Effective Surface Radius of Thickness/ Index of Abbe Aperture No. Element Curvature Distance Refraction Number Radius S1 First 1.9512 0.4488 1.546 56.114 1.307 S2 Lens 3.1152 0.1260 1.253 S3 Second 2.8686 0.5753 1.546 56.114 1.214 S4 Lens

-12.9825 0.0186 1.180 S5 Third Lens 4.5064 0.1856 1.679 19.236 1.074 (Stop) S6 2.1969
0.5197 1.016 S7 Fourth -2108.865 0.2796 1.679 19.236 1.179 S8 Lens -6755.436 0.1715 1.338
S9 Fifth 3.1135 0.2734 1.546 56.114 1.528 S10 Lens 3.2672 0.2417 1.808 S11 Sixth 3.2228
0.3650 1.679 19.236 1.996 S12 Lens 2.5388 0.1438 2.320 S13 Seventh 1.4451 0.5122 1.537
53.955 2.500 S14 Lens 1.2680 0.2501 2.738 S15 Filter Infinity 0.1100 1.519 64.166 2.940 S16
Infinity 0.5924 2.971 S17 Imaging Infinity 0.0054 3.251 Plane
TABLE-US-00008 TABLE 8 K A B C D E F G H J S1 -7.5279 0.0857 -0.105 0.0528 -0.0256
-0.0221 0.0379 -0.0166 0.0023 0 S2 -19.893 -0.0142 -0.1337 0.0682 0.0621 -0.0783 0.0306
-0.0031 -0.0006 0 S3 -0.0142 -0.0449 -0.0418 -0.0147 0.1136 0.012 -0.1333 0.0892 -0.0193 0
S4 0 0.0281 -0.189 0.276 -0.0808 -0.2297 0.2908 -0.1382 0.024 0 S5 -6.2325 -0.0763 -0.0054
-0.0795 0.6054 -1.1875 1.107 -0.5047 0.0912 0 S6 0.4782 -0.115 0.1396 -0.2676 0.5637
-0.7991 0.6898 -0.325 0.0682 0 S7 0 -0.0188 -0.0772 0.0717 0.0184 -0.081 0.0225 0.0277
-0.0139 0 S8 0 -0.0127 -0.1356 0.0837 0.0781 -0.1502 0.0847 -0.0163 0 0 S9 -49.08 0.1815
-0.3205 0.2837 -0.2161 0.1317 -0.0595 0.0158 -0.0017 0 S10 -5.4303 -0.0205 0.025 -0.1003
0.1046 -0.0624 0.0222 -0.0043 0.0003 0 S11 -1.136 0.0314 -0.2615 0.3261 -0.2695 0.133
-0.0369 0.0053 -0.0003 0 S12 0.0272 -0.1293 0.0241 5E-05 -0.0123 0.0085 -0.0024 0.0003
-2E-05 0 S13 -0.8 -0.5247 0.2994 -0.1227 0.0414 -0.0108 0.002 -0.0002 2E-05 -4E-07 S14
-1.3207 -0.3666 0.2425 -0.1248 0.0468 -0.0121 0.002 -0.0002 1E-05 -3E-07

Fifth Example

[0171] FIG. 9 is a view illustrating a fifth example of an optical imaging system, and FIG. 10 illustrates aberration curves of the optical imaging system of FIG. 9.

[0172] An optical imaging system 5 includes a first lens 1005, a second lens 2005, a third lens 3005, a fourth lens 4005, a fifth lens 5005, a sixth lens 6005, and a seventh lens 7005.

[0173] The first lens 1005 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens 2005 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens 3005 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens 4005 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens 5005 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens 6005 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens 6005. The seventh lens 7005 has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens 7005.

[0174] The optical imaging system 5 further includes a stop, a filter 8005, and an image sensor 9005. The stop is disposed between the first lens 1005 and the second lens 2005 to adjust an amount of light incident on the image sensor 9005. The filter 8005 is disposed between the seventh lens 7005 and the image sensor 9005 to block infrared rays. The image sensor 9005 forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 9, the stop is disposed at a distance of 0.383 mm from the object-side surface of the first lens 1005 toward the imaging plane of the optical imaging system 5. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 5 listed in Table 59 that appears later in this application.

[0175] Table 9 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 9, and Table 10 below shows aspherical surface coefficients of the lenses of FIG. 9.

TABLE-US-00009 TABLE 9 Radius Thick- Index of Effective Surface of ness/ Re- Abbe Aperture

No. Element Curvature Distance fraction Number Radius S1 First 2.1824 0.3329 1.546 56.114
 1.380 S2 Lens 1.9439 0.0500 1.369 S3 (Stop) Second 1.6857 0.7322 1.546 56.114 1.335 S4 Lens
 28.3727 0.0500 1.264 S5 Third 7.1536 0.2200 1.679 19.236 1.185 S6 Lens 2.9223 0.4264 1.050 S7
 Fourth 46.9146 0.3121 1.646 23.528 1.112 S8 Lens 17.5860 0.2616 1.268 S9 Fifth 2.2655 0.2700
 1.646 23.528 1.774 S10 Lens 2.3143 0.3731 1.839 S11 Sixth 8.5186 0.6078 1.546 56.114 2.160
 S12 Lens -1.9871 0.3782 2.308 S13 Seventh -4.7165 0.3600 1.546 56.114 2.780 S14 Lens 1.8919
 0.1457 2.998 S15 Filter Infinity 0.1100 1.519 64.166 3.353 S16 Infinity 0.6600 3.385 S17 Imaging
 Infinity 0.0100 3.712 Plane

TABLE-US-00010 TABLE 10 K A B C D E F G H J S1 -3.5715 0.0005 0.0011 -0.0181 0.0025
 0.0107 -0.0084 0.0026 -0.0003 0 S2 -9.1496 -0.0513 -0.0055 0.0116 0.0161 -0.0207 0.0078
 -0.001 0 0 S3 -2.5622 -0.0879 0.1115 -0.1204 0.1625 -0.1325 0.0578 -0.0118 0.0006 0 S4 -90
 -0.078 0.2103 -0.4384 0.6397 -0.6153 0.3736 -0.1288 0.0189 0 S5 0 -0.1133 0.2975 -0.5447
 0.7496 -0.7199 0.4525 -0.1642 0.0257 0 S6 4.6946 -0.0705 0.1434 -0.2144 0.1998 -0.0956
 -0.0142 0.0399 -0.0137 0 S7 0 -0.0972 0.1221 -0.3303 0.5457 -0.6222 0.4555 -0.1995 0.0405 0
 S8 0 -0.1596 0.2027 -0.3281 0.3412 -0.2472 0.1212 -0.0385 0.0064 0 S9 -18.27 -0.0564
 -0.0069 0.0518 -0.0566 0.0228 -0.0011 -0.0019 0.0004 0 S10 -15.127 -0.0603 -0.0145 0.0594
 -0.0601 0.0318 -0.0096 0.0015 -1E-04 0 S11 0 0.0027 -0.0398 0.025 -0.0137 0.005 -0.001
 1E-04 -4E-06 0 S12 -1.1693 0.1224 -0.1006 0.0535 -0.0195 0.005 -0.0008 8E-05 -3E-06 0
 S13 -4.4446 -0.097 -0.0137 0.0358 -0.0141 0.0028 -0.0003 2E-05 -5E-07 0 S14 -8.7431
 -0.0906 0.0342 -0.009 0.0017 -0.0002 2E-05 -1E-06 3E-08 0

Sixth Example

[0176] FIG. 11 is a view illustrating a sixth example of an optical imaging system, and FIG. 12 illustrates aberration curves of the optical imaging system of FIG. 11.

[0177] An optical imaging system 6 includes a first lens 1006, a second lens 2006, a third lens 3006, a fourth lens 4006, a fifth lens 5006, a sixth lens 6006, and a seventh lens 7006.

[0178] The first lens 1006 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens 2006 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens 3006 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens 4006 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens 5006 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens 6006 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens 6006. The seventh lens 7006 has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens 7006.

[0179] The optical imaging system 6 further includes a stop, a filter 8006, and an image sensor 9006. The stop is disposed between the first lens 1006 and the second lens 2006 to adjust an amount of light incident on the image sensor 9006. The filter 8006 is disposed between the seventh lens 7006 and the image sensor 9006 to block infrared rays. The image sensor 9006 forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 11, the stop is disposed at a distance of 0.352 mm from the object-side surface of the first lens 1006 toward the imaging plane of the optical imaging system 6. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 6 listed in Table 59 that appears later in this application.

[0180] Table 11 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 11, and Table 12 below shows aspherical surface coefficients of the lenses

of FIG. 11.

TABLE-US-00011 TABLE 11 Radius Thick- Index Effective Surface of ness/ of Re- Abbe
Aperture No. Element Curvature Distance fraction Number Radius S1 First 2.0068 0.3061 1.546
56.114 1.269 S2 Lens 1.7875 0.0460 1.259 S3 (Stop) Second 1.5501 0.6732 1.546 56.114 1.228 S4
Lens 26.0899 0.0460 1.162 S5 Third 6.5780 0.2023 1.679 19.236 1.090 S6 Lens 2.6872 0.3921
0.966 S7 Fourth 43.1399 0.2870 1.646 23.528 1.017 S8 Lens 16.1710 0.2406 1.159 S9 Fifth
2.0832 0.2483 1.646 23.528 1.449 S10 Lens 2.1281 0.3430 1.675 S11 Sixth 7.8332 0.5589 1.546
56.114 1.986 S12 Lens -1.8272 0.3478 2.083 S13 Seventh -4.3370 0.3310 1.546 56.114 2.556 S14
Lens 1.7397 0.1457 2.669 S15 Filter Infinity 0.1100 1.519 64.166 2.953 S16 Infinity 0.5893 2.983
S17 Imaging Infinity 0.0100 3.258 Plane

TABLE-US-00012 TABLE 12 K A B C D E F G H J S1 -3.5715 0.0006 0.0017 -0.0326 0.0053
0.027 -0.0251 0.0092 -0.0013 0 S2 -9.1496 -0.0659 -0.0083 0.0208 0.0344 -0.0521 0.0231
-0.0036 0 0 S3 -2.5622 -0.113 0.1695 -0.2165 0.3458 -0.3334 0.1719 -0.0415 0.0024 0 S4 -90
-0.1003 0.3198 -0.7886 1.361 -1.5481 1.1116 -0.4533 0.0785 0 S5 0 -0.1458 0.4526 -0.9799
1.5947 -1.8113 1.3465 -0.5778 0.107 0 S6 4.6946 -0.0906 0.2181 -0.3857 0.425 -0.2404
-0.0421 0.1403 -0.0569 0 S7 0 -0.125 0.1857 -0.5941 1.161 -1.5655 1.3554 -0.7021 0.1686 0
S8 0 -0.2053 0.3083 -0.5902 0.7259 -0.622 0.3607 -0.1355 0.0267 0 S9 -18.27 -0.0725 -0.0104
0.0931 -0.1203 0.0574 -0.0031 -0.0068 0.0017 0 S10 -15.127 -0.0776 -0.022 0.1069 -0.1279
0.0799 -0.0284 0.0054 -0.0004 0 S11 0 0.0035 -0.0606 0.045 -0.0292 0.0125 -0.0029 0.0003
-2E-05 0 S12 -1.1693 0.1574 -0.153 0.0962 -0.0415 0.0126 -0.0025 0.0003 -1E-05 0 S13
-4.4446 -0.1248 -0.0209 0.0644 -0.0301 0.007 -0.0009 6E-05 -2E-06 0 S14 -8.7431 -0.1165
0.052 -0.0161 0.0035 -0.0006 7E-05 -4E-06 1E-07 0

Seventh Example

[0181] FIG. 13 is a view illustrating a seventh example of an optical imaging system, and FIG. 14 illustrates aberration curves of the optical imaging system of FIG. 13.

[0182] An optical imaging system 7 includes a first lens 1007, a second lens 2007, a third lens 3007, a fourth lens 4007, a fifth lens 5007, a sixth lens 6007, and a seventh lens 7007.

[0183] The first lens 1007 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens 2007 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens 3007 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens 4007 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens 5007 has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The sixth lens 6007 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens 6007. The seventh lens 7007 has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens 7007.

[0184] The optical imaging system 7 further includes a stop, a filter 8007, and an image sensor 9007. The stop is disposed between the first lens 1007 and the second lens 2007 to adjust an amount of light incident on the image sensor 9007. The filter 8007 is disposed between the seventh lens 7007 and the image sensor 9007 to block infrared rays. The image sensor 9007 forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 13, the stop is disposed at a distance of 0.731 mm from the object-side surface of the first lens 1007 toward the imaging plane of the optical imaging system 7. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 7 listed in Table 59 that appears later in this application.

[0185] Table 13 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 13, and Table 14 below shows aspherical surface coefficients of the lenses of FIG. 13.

TABLE-US-00013 TABLE 13 Radius Thick- Index Effective Surface of ness/ of Re- Abbe
Aperture No. Element Curvature Distance fraction Number Radius S1 First 1.73233 0.73124 1.546
56.114 1.250 S2 (Stop) Lens 12.53699 0.07002 1.181 S3 Second 5.58930 0.20000 1.667 20.353
1.147 S4 Lens 2.57397 0.39715 1.100 S5 Third 8.06552 0.38474 1.546 56.114 1.128 S6 Lens
7.83668 0.19259 1.247 S7 Fourth 6.68716 0.24423 1.546 56.114 1.276 S8 Lens 30.32847 0.27130
1.374 S9 Fifth -3.28742 0.24968 1.667 20.353 1.481 S10 Lens -4.51593 0.13884 1.734 S11 Sixth
5.67988 0.51987 1.546 56.114 2.150 S12 Lens -1.89003 0.31663 2.318 S13 Seventh -3.93255
0.30000 1.546 56.114 2.640 S14 Lens 1.74183 0.19371 2.747 S15 Filter Infinity 0.11000 1.518
64.166 3.146 S16 Infinity 0.77000 3.177 S17 Imaging Infinity 0.01000 3.536 Plane

TABLE-US-00014 TABLE 14 K A B C D E F G H J S1 -0.7464 0.0139 0.0344 -0.0749 0.1029
-0.0706 0.0173 0.0042 -0.0023 0 S2 36.669 -0.0823 0.195 -0.3067 0.3634 -0.323 0.1902
-0.0632 0.0086 0 S3 -1.3559 -0.1603 0.3305 -0.4059 0.3324 -0.1787 0.0673 -0.0166 0.0018 0
S4 -0.4109 -0.0907 0.1444 0.1155 -0.7969 1.5009 -1.4406 0.7219 -0.147 0 S5 0 -0.0739 0.0463
-0.1203 0.1165 -0.0578 -0.0089 0.0233 -0.0057 0 S6 0 -0.0932 0.0034 0.0521 -0.1827 0.2457
-0.2173 0.1126 -0.0241 0 S7 25.148 -0.1235 -0.1887 0.3763 -0.554 0.6731 -0.5796 0.2782
-0.0538 0 S8 -99 -9E-05 -0.3274 0.3588 -0.3195 0.3451 -0.2608 0.0995 -0.0144 0 S9 -70.894
0.0205 0.0483 -0.5284 0.7583 -0.4915 0.1636 -0.0271 0.0018 0 S10 2.2832 0.1759 -0.3448
0.2283 -0.0716 0.011 -0.0007 -4E-06 1E-06 0 S11 -99 0.1188 -0.2169 0.1675 -0.0871 0.0276
-0.0049 0.0005 -2E-05 0 S12 -3.3067 0.1644 -0.1849 0.1159 -0.049 0.0138 -0.0024 0.0002
-9E-06 0 S13 -2.4772 -0.1026 -0.0482 0.074 -0.0308 0.0067 -0.0008 6E-05 -2E-06 0 S14
-1.1028 -0.2935 0.2033 -0.1127 0.0457 -0.0129 0.0024 -0.0003 2E-05 -5E-07

Eighth Example

[0186] FIG. 15 is a view illustrating an eighth example of an optical imaging system, and FIG. 16 illustrates aberration curves of the optical imaging system of FIG. 15.

[0187] An optical imaging system 8 includes a first lens 1008, a second lens 2008, a third lens 3008, a fourth lens 4008, a fifth lens 5008, a sixth lens 6008, and a seventh lens 7008.

[0188] The first lens 1008 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens 2008 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens 3008 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens 4008 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens 5008 has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The sixth lens 6008 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens 6008. The seventh lens 7008 has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens 7008.

[0189] The optical imaging system 8 further includes a stop, a filter 8008, and an image sensor 9008. The stop is disposed between the first lens 1008 and the second lens 2008 to adjust an amount of light incident on the image sensor 9008. The filter 8008 is disposed between the seventh lens 7008 and the image sensor 9008 to block infrared rays. The image sensor 9008 forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 15, the stop is disposed at a distance of 0.696 mm from the object-side surface of the first lens 1008 toward the imaging plane of the optical imaging system 8. This distance is equal to TTL-SL and can be

calculated from the values of TTL and SL for Example 8 listed in Table 59 that appears later in this application.

[0190] Table 15 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 15, and Table 16 below shows aspherical surface coefficients of the lenses of FIG. 15.

TABLE-US-00015 TABLE 15 Radius Thick- Index Effective Surface of ness/ of Re- Abbe
Aperture No. Element Curvature Distance fraction Number Radius S1 First 1.76093 0.69658 1.546
56.114 1.285 S2 (Stop) Lens 12.78634 0.09190 1.237 S3 Second 5.69283 0.19245 1.667 20.353
1.198 S4 Lens 2.65937 0.39218 1.100 S5 Thir 8.20428 0.41830 1.546 56.114 1.143 S6 Lens
7.91545 0.16876 1.265 S7 Fourth 6.75201 0.26903 1.546 56.114 1.284 S8 Lens 55.74455 0.23097
1.367 S9 Fifth -3.08690 0.42919 1.667 20.353 1.418 S10 Lens -4.62560 0.09404 1.699 S11 Sixth
5.18773 0.58281 1.546 56.114 2.150 S12 Lens -1.85438 0.30690 2.100 S13 Seventh -4.06715
0.30000 1.546 56.114 2.300 S14 Lens 1.68940 0.22233 2.631 S15 Filter Infinity 0.11000 1.518
64.166 2.912 S16 Infinity 0.77152 2.942 S17 Imaging Infinity 0.00847 3.281 Plane

TABLE-US-00016 TABLE 16 K A B C D E F G H J S1 -0.785 0.0169 0.0171 -0.0159 -0.0114
0.0583 -0.067 0.0338 -0.0066 0 S2 45.58 -0.0601 0.1355 -0.2175 0.2636 -0.2331 0.1319
-0.0411 0.0051 0 S3 1.584 -0.1444 0.2883 -0.3594 0.3009 -0.1422 0.021 0.0121 -0.0046 0 S4
-0.5948 -0.0935 0.1656 -0.0201 -0.372 0.7699 -0.7293 0.3536 -0.0683 0 S5 0 -0.0677 0.0256
-0.0658 0.0582 -0.0383 0.013 0.0013 -0.0005 0 S6 0 -0.0847 -0.0499 0.1996 -0.3976 0.4333
-0.3115 0.1361 -0.026 0 S7 25.588 -0.1098 -0.2727 0.6279 -0.9136 0.9353 -0.6674 0.2828
-0.0509 0 S8 -99 0.0297 -0.4755 0.7447 -0.8912 0.8375 -0.5032 0.1625 -0.0211 0 S9 -66.103
-0.0453 0.1301 -0.541 0.7178 -0.4567 0.1508 -0.0248 0.0016 0 S10 1.6574 0.126 -0.2272
0.1439 -0.0478 0.0097 -0.0013 0.0001 -5E-06 0 S11 -99 0.1115 -0.1711 0.1182 -0.0561 0.0162
-0.0026 0.0002 -8E-06 0 S12 -3.3351 0.1192 -0.1147 0.0615 -0.0222 0.0053 -0.0008 6E-05
-2E-06 0 S13 -2.7152 -0.0748 -0.0688 0.08 -0.032 0.0069 -0.0009 6E-05 -2E-06 0 S14
-1.1007 -0.2576 0.1553 -0.0741 0.0262 -0.0065 0.0011 -0.0001 7E-06 -2E-07

Ninth Example

[0191] FIG. 17 is a view illustrating a ninth example of an optical imaging system, and FIG. 18 illustrates aberration curves of the optical imaging system of FIG. 17.

[0192] An optical imaging system 9 includes a first lens 1009, a second lens 2009, a third lens 3009, a fourth lens 4009, a fifth lens 5009, a sixth lens 6009, and a seventh lens 7009.

[0193] The first lens 1009 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens 2009 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. The third lens 3009 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens 4009 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens 5009 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens 6009 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens 6009. The seventh lens 7009 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens 7009, and one inflection point is formed on the image-side surface of the seventh lens 7009.

[0194] The optical imaging system 9 further includes a stop, a filter 8009, and an image sensor 9009. The stop is disposed between the second lens 2009 and the third lens 3009 to adjust an amount of light incident on the image sensor 9009. The filter 8009 is disposed between the seventh lens 7009 and the image sensor 9009 to block infrared rays. The image sensor 9009 forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 17, the

stop is disposed at a distance of 1.158 mm from the object-side surface of the first lens **1009** toward the imaging plane of the optical imaging system **9**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 9 listed in Table 59 that appears later in this application.

[0195] Table 17 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **17**, and Table 18 below shows aspherical surface coefficients of the lenses of FIG. **17**.

TABLE-US-00017 TABLE 17 Radius Thick- Index Effective Surface of ness/ of Re- Abbe Aperture No. Element Curvature Distance fraction Number Radius S1 First 2.141 0.481 1.546 56.114 1.450 S2 Lens 3.251 0.110 1.350 S3 Second 3.253 0.542 1.546 56.114 1.285 S4 Lens -15.773 0.025 1.232 S5 (Stop) Third 8.425 0.230 1.679 19.236 1.157 S6 Lens 3.514 0.625 1.095 S7 Fourth 25.986 0.296 1.679 19.236 1.265 S8 Lens 15.894 0.230 1.452 S9 Fifth 3.048 0.400 1.546 56.114 1.675 S10 Lens 3.616 0.290 2.092 S11 Sixth 3.762 0.400 1.679 19.236 2.153 S12 Lens 2.792 0.204 2.476 S13 Seventh 1.614 0.510 1.537 53.955 2.938 S14 Lens 1.326 0.196 3.102 S15 Filter Infinity 0.110 1.518 64.197 3.420 S16 Infinity 0.639 3.450 S17 Imaging Infinity 0.011 3.730 Plane

TABLE-US-00018 TABLE 18 K A B C D E F G H J S1 -8.038 0.0707 -0.0797 0.0334 0.0072 -0.0491 0.0465 -0.0186 0.0032 -0.0002 S2 -20.594 -0.0019 -0.1494 0.2041 -0.2922 0.3755 -0.3085 0.1486 -0.0387 0.0042 S3 -0.0908 -0.0339 -0.0641 0.1368 -0.2821 0.4921 -0.4815 0.2605 -0.0746 0.0088 S4 -0.4822 -0.0436 0.1761 -0.3256 0.1999 0.1916 -0.4291 0.3203 -0.1141 0.0162 S5 -1.1841 -0.1073 0.2544 -0.4683 0.4991 -0.2863 0.0565 0.0325 -0.0229 0.0044 S6 0.8733 -0.0693 0.0357 0.2048 -0.8833 1.7328 -1.9742 1.3464 -0.5106 0.083 S7 -0.4999 -0.0314 0.0135 -0.2894 0.9716 -1.7181 1.7923 -1.1152 0.3837 -0.0563 S8 -1E-06 -0.0273 -0.1177 0.212 -0.2544 0.2157 -0.1264 0.0469 -0.0093 0.0007 S9 -41.843 0.1624 -0.3487 0.4016 -0.3105 0.1396 -0.027 -0.0038 0.0026 -0.0003 S10 -5.1424 0.0397 -0.1364 0.1569 -0.1229 0.0633 -0.0212 0.0044 -0.0005 3E-05 S11 -2.1666 0.0056 -0.1809 0.1985 -0.1438 0.0641 -0.0173 0.0028 -0.0002 9E-06 S12 -0.0207 -0.1043 0.0239 -0.0063 -0.0007 0.0007 -3E-06 -4E-05 7E-06 -4E-07 S13 -0.7948 -0.4128 0.1863 -0.0516 0.0101 -0.0015 0.0002 -1E-05 6E-07 -1E-08 S14 -1.3226 -0.3105 0.1713 -0.0712 0.0213 -0.0043 0.0006 -5E-05 2E-06 -5E-08

Tenth Example

[0196] FIG. **19** is a view illustrating a tenth example of an optical imaging system, and FIG. **20** illustrates aberration curves of the optical imaging system of FIG. **19**.

[0197] An optical imaging system **10** includes a first lens **1010**, a second lens **2010**, a third lens **3010**, a fourth lens **4010**, a fifth lens **5010**, a sixth lens **6010**, and a seventh lens **7010**.

[0198] The first lens **1010** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2010** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3010** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4010** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5010** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6010** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6010**. The seventh lens **7010** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens **7010**, and one inflection point is formed on the image-side surface of the seventh lens **7010**.

[0199] The optical imaging system **10** further includes a stop, a filter **8010**, and an image sensor

9010. The stop is disposed between the second lens **2010** and the third lens **3010** to adjust an amount of light incident on the image sensor **9010**. The filter **8010** is disposed between the seventh lens **7010** and the image sensor **9010** to block infrared rays. The image sensor **9010** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **19**, the stop is disposed at a distance of 1.179 mm from the object-side surface of the first lens **1010** toward the imaging plane of the optical imaging system **10**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 10 listed in Table 59 that appears later in this application.

[0200] Table 19 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **19**, and Table 20 below shows aspherical surface coefficients of the lenses of FIG. **19**.

TABLE-US-00019 TABLE 19 Thick- Index Effective Surface Radius of ness/ of Re- Abbe Aperture No. Element Curvature Distance fraction Number Radius S1 First 2.3878602 0.35 1.546 56.114 1.470 S2 Lens 2.62132398 0.10425 1.439 S3 Second 2.53822644 0.7 1.546 56.114 1.405 S4 Lens 36.207255 0.025 1.322 S5 (Stop) Third 7.79607111 0.23 1.679 19.236 1.287 S6 Lens 3.49120755 0.3961335 1.325 S7 Fourth 5.43312406 0.5053863 1.546 56.114 1.461 S8 Lens 9.39050621 0.4436731 1.563 S9 Fifth 4.67888662 0.4842217 1.546 56.114 1.772 S10 Lens 9.57328318 0.4757916 2.209 S11 Sixth 7.18942036 0.4753531 1.679 19.236 2.238 S12 Lens 3.83971065 0.2154454 2.557 S13 Seventh 1.81373827 0.5598542 1.546 56.114 3.026 S14 Lens 1.38723655 0.2300778 3.262 S15 Filter Infinity 0.11 1.518 64.197 3.724 S16 Infinity 0.6349996 3.763 S17 Imaging Infinity 0.015 4.160 Plane

TABLE-US-00020 TABLE 20 K A B C D E F G H J S1 -0.9977 -0.0046 -0.0139 -0.024 0.0712 -0.0925 0.0658 -0.0257 0.0051 -0.0004 S2 -10.008 0.0879 -0.2624 0.4378 -0.5581 0.4751 -0.2476 0.0753 -0.0122 0.0008 S3 -0.5121 -0.0082 0.0297 -0.2049 0.402 -0.4509 0.3219 -0.1392 0.0325 -0.0031 S4 42.587 -0.0077 -0.0816 0.5107 -1.169 1.3206 -0.8193 0.2839 -0.0514 0.0038 S5 14.891 -0.0724 0.0409 0.4245 -1.2505 1.5731 -1.0799 0.4229 -0.0892 0.0079 S6 1.8054 -0.0886 0.1906 -0.2777 0.2837 -0.2167 0.1131 -0.0327 0.0033 0.0002 S7 -10.152 0.028 -0.4038 1.2639 -2.2626 2.4845 -1.6998 0.7029 -0.16 0.0153 S8 1.7534 -0.0643 0.1117 -0.3935 0.7438 -0.8306 0.565 -0.2306 0.0519 -0.005 S9 -44.62 0.0891 -0.1333 0.115 -0.0807 0.043 -0.017 0.0044 -0.0006 4E-05 S10 -4.9001 0.0976 -0.137 0.0999 -0.0465 0.0132 -0.0022 0.0002 -1E-05 2E-07 S11 -13.159 0.0954 -0.1535 0.1202 -0.0703 0.0264 -0.006 0.0008 -6E-05 2E-06 S12 0.7792 -0.0211 -0.0288 0.0188 -0.0093 0.003 -0.0006 6E-05 -4E-06 9E-08 S13 -0.8786 -0.3003 0.0918 -0.0015 -0.0073 0.0023 -0.0004 3E-05 -1E-06 3E-08 S14 -1.3086 -0.2504 0.1203 -0.0427 0.0114 -0.0021 0.0003 -2E-05 9E-07 -2E-08

Eleventh Example

[0201] FIG. **21** is a view illustrating an eleventh example of an optical imaging system, and FIG. **22** illustrates aberration curves of the optical imaging system of FIG. **21**.

[0202] An optical imaging system **11** includes a first lens **1011**, a second lens **2011**, a third lens **3011**, a fourth lens **4011**, a fifth lens **5011**, a sixth lens **6011**, and a seventh lens **7011**.

[0203] The first lens **1011** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2011** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3011** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4011** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5011** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6011** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6011**. The seventh lens **7011** has a negative refractive power,

and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens **7011**, and one inflection point is formed on the image-side surface of the seventh lens **7011**.

[0204] The optical imaging system **11** further includes a stop, a filter **8011**, and an image sensor **9011**. The stop is disposed between the second lens **2011** and the third lens **3011** to adjust an amount of light incident on the image sensor **9011**. The filter **8011** is disposed between the seventh lens **7011** and the image sensor **9011** to block infrared rays. The image sensor **9011** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **21**, the stop is disposed at a distance of 1.205 mm from the object-side surface of the first lens **1011** toward the imaging plane of the optical imaging system **11**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 11 listed in Table 59 that appears later in this application.

[0205] Table 21 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **21**, and Table 22 below shows aspherical surface coefficients of the lenses of FIG. **21**.

TABLE-US-00021 TABLE 21 Index Effective Surface Radius of Thickness/ of Re- Abbe Aperture No. Element Curvature Distance fraction Number Radius S1 First 2.34218008 0.35 1.546 56.114 1.570 S2 Lens 2.46082267 0.130273 1.582 S3 Second 2.30906902 0.7 1.546 56.114 1.578 S4 Lens 38.8335574 0.025 1.553 S5 (Stop) Third 7.6526629 0.2529802 1.679 19.236 1.474 S6 Lens 3.32721743 0.5047159 1.325 S7 Fourth 5.67437537 0.3466492 1.679 19.236 1.461 S8 Lens 6.58110754 0.40646 1.563 S9 Fifth 4.79255093 0.35 1.546 56.114 1.772 S10 Lens 10.114446 0.5278215 2.209 S11 Sixth 6.16226916 0.43 1.679 19.236 2.238 S12 Lens 3.92995117 0.2624604 2.557 S13 Seventh 1.96658005 0.5643968 1.546 56.114 3.026 S14 Lens 1.49045699 0.2268426 3.262 S15 Filter Infinity 0.11 1.518 64.197 3.641 S16 Infinity 0.7973487 3.681 S17 Imaging Infinity 0.015 4.171 Plane

TABLE-US-00022 TABLE 22 K A B C D E F G H J S1 -1 -0.0146 0.01 -0.0522 0.0643 -0.0468 0.0217 -0.0061 0.0009 -6E-05 S2 -9.9316 0.0529 -0.0805 0.0187 0.0015 0.0183 -0.0206 0.0089 -0.0018 0.0001 S3 -0.3035 -0.0312 0.1094 -0.2703 0.3068 -0.1783 0.0505 -0.0026 -0.0019 0.0003 S4 42.587 -0.14 0.5905 -1.3419 1.845 -1.6056 0.8858 -0.299 0.0561 -0.0045 S5 14.878 -0.1698 0.5798 -1.2212 1.6455 -1.4398 0.807 -0.2778 0.0533 -0.0044 S6 2.4782 -0.0974 0.3302 -0.8477 1.4713 -1.6606 1.1916 -0.5238 0.1288 -0.0136 S7 -10.89 -0.0126 -0.1352 0.3555 -0.5088 0.426 -0.2043 0.0479 -0.0019 -0.0008 S8 2.4559 -0.0438 -0.0219 -0.0159 0.1351 -0.2288 0.1952 -0.0929 0.0235 -0.0025 S9 -44.519 0.0902 -0.1408 0.1256 -0.0754 0.0253 -0.0032 -0.0008 0.0003 -3E-05 S10 2.7864 0.0708 -0.1198 0.1033 -0.057 0.0193 -0.004 0.0005 -3E-05 9E-07 S11 -17.823 0.0653 -0.1084 0.075 -0.0394 0.0131 -0.0026 0.0003 -2E-05 5E-07 S12 0.5903 -0.0273 -0.0096 0.0039 -0.0025 0.001 -0.0002 2E-05 -1E-06 3E-08 S13 -0.8439 -0.2851 0.1057 -0.0237 0.0034 -0.0003 5E-07 2E-06 -1E-07 4E-09 S14 -1.3672 -0.238 0.119 -0.0473 0.0138 -0.0027 0.0003 -3E-05 1E-06 -2E-08

Twelfth Example

[0206] FIG. **23** is a view illustrating a twelfth example of an optical imaging system, and FIG. **24** illustrates aberration curves of the optical imaging system of FIG. **23**.

[0207] An optical imaging system **12** includes a first lens **1012**, a second lens **2012**, a third lens **3012**, a fourth lens **4012**, a fifth lens **5012**, a sixth lens **6012**, and a seventh lens **7012**.

[0208] The first lens **1012** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2012** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3012** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4012** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5012** has a negative refractive power, and an object-side surface thereof is concave

and an image-side surface thereof is convex. The sixth lens **6012** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6012**. The seventh lens **7012** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens **7012**.

[0209] The optical imaging system **12** further includes a stop, a filter **8012**, and an image sensor **9012**. The stop is disposed in front of the first lens **1012** to adjust an amount of light incident on the image sensor **9012**. The filter **8012** is disposed between the seventh lens **7012** and the image sensor **9012** to block infrared rays. The image sensor **9012** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **23**, the stop is disposed at a distance of 0.250 mm from the object-side surface of the first lens **1012** toward the imaging plane of the optical imaging system **12**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 12 listed in Table 59 that appears later in this application.

[0210] Table 23 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **23**, and Table 24 below shows aspherical surface coefficients of the lenses of FIG. **23**.

TABLE-US-00023 TABLE 23 Index of Effective Surface Radius of Thickness/ Refrac- Abbe Aperture No. Element Curvature Distance tion Number Radius S1 First 1.7211 0.6349 1.544 56.114 1.100 (Stop) Lens S2 11.4571 0.1212 1.071 S3 Second 119.1721 0.2033 1.661 20.353 1.057 S4 Lens 4.4758 0.0843 1.043 S5 Third 4.5258 0.3109 1.544 56.114 1.051 S6 Lens 20.6082 0.2158 1.015 S7 Fourth 13.2152 0.2369 1.544 56.114 1.019 S8 Lens 16.2733 0.2103 1.070 S9 Fifth -6.5732 0.4119 1.651 21.494 1.076 S10 Lens -10.4553 0.3710 1.320 S11 Sixth 3.4779 0.6318 1.544 56.114 1.556 S12 Lens 3.1994 0.2672 2.337 S13 Seventh 2.8804 0.5060 1.544 56.114 2.489 S14 Lens 1.7054 0.1384 2.666 S15 Filter Infinity 0.2100 3.102 S16 Infinity 0.5794 3.177 S17 Imaging Infinity 0.0106 3.529 Plane

TABLE-US-00024 TABLE 24 K A B C D E F G H S1 0.0432 -0.0088 0.0131 -0.0627 0.1199 -0.1345 0.077 -0.018 -0.0004 S2 -26.097 -0.0562 0.051 -0.0514 0.0595 -0.0683 0.0462 -0.0139 -7E-05 S3 -99 -0.1283 0.1953 -0.2779 0.5135 -0.8812 0.9662 -0.5723 0.1395 S4 -16.567 -0.0971 0.1552 -0.3608 0.985 -2.059 2.5647 -1.6683 0.4378 S5 -1.6774 -0.0377 0.065 -0.4515 1.687 -3.5163 4.2391 -2.6607 0.6752 S6 57.913 -0.0559 0.0533 -0.341 1.3373 -2.8539 3.4811 -2.2114 0.5781 S7 -66.305 -0.1749 -0.0635 0.0963 -0.2061 0.5819 -0.9 0.6874 -0.1979 S8 19.549 -0.1228 -0.0686 0.0207 0.1647 -0.2695 0.1725 -0.0616 0.0161 S9 29.709 -0.0709 0.0826 -0.3062 0.6009 -0.6459 0.3344 -0.0761 0 S10 -31.338 -0.1255 0.1076 -0.1494 0.1908 -0.1423 0.0506 -0.0065 0 S11 -46.453 0.0038 -0.1455 0.1534 -0.126 0.0705 -0.0225 0.0029 0 S12 -31.504 0.0093 -0.0326 0.0149 -0.0033 0.0003 -1E-05 -7E-07 0 S13 -0.5233 -0.2947 0.1709 -0.0627 0.0154 -0.0025 0.0003 -1E-05 3E-07 S14 -0.8257 -0.2584 0.1353 -0.0565 0.0166 -0.0032 0.0004 -3E-05 7E-07

Thirteenth Example

[0211] FIG. **25** is a view illustrating a thirteenth example of an optical imaging system, and FIG. **26** illustrates aberration curves of the optical imaging system of FIG. **25**.

[0212] An optical imaging system **13** includes a first lens **1013**, a second lens **2013**, a third lens **3013**, a fourth lens **4013**, a fifth lens **5013**, a sixth lens **6013**, and a seventh lens **7013**.

[0213] The first lens **1013** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2013** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3013** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4013** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave.

The fifth lens **5013** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The sixth lens **6013** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6013**. The seventh lens **7013** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens **7013**, and one inflection point is formed on the image-side surface of the seventh lens **7013**.

[0214] The optical imaging system **13** further includes a stop, a filter **8013**, and an image sensor **9013**. The stop is disposed in front of the first lens **1013** to adjust an amount of light incident on the image sensor **9013**. The filter **8013** is disposed between the seventh lens **7013** and the image sensor **9013** to block infrared rays. The image sensor **9013** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 25, the stop is disposed at a distance of 0.200 mm from the object-side surface of the first lens **1013** toward the imaging plane of the optical imaging system **13**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 13 listed in Table 59 that appears later in this application.

[0215] Table 25 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 25, and Table 26 below shows aspherical surface coefficients of the lenses of FIG. 25.

TABLE-US-00025 TABLE 25 Index of Effective Surface Radius of Thickness/ Refrac- Abbe Aperture No. Element Curvature Distance tion Number Radius S1 First 2.0468 0.6797 1.544 56.114 1.100 (Stop) Lens S2 15.4382 0.1358 1.127 S3 Second 5.7788 0.2300 1.661 20.353 1.142 S4 Lens 2.6751 0.2609 1.153 S5 Third 4.0817 0.2952 1.544 56.114 1.246 S6 Lens 7.3687 0.3995 1.258 S7 Fourth 7.0190 0.3141 1.544 56.114 1.340 S8 Lens 7.7267 0.2822 1.388 S9 Fifth -9.0899 0.3555 1.661 20.353 1.405 S10 Lens -9.9293 0.1000 1.576 S11 Sixth 3.0002 0.6701 1.544 56.114 1.700 S12 Lens 3.7492 0.3200 2.316 S13 Seventh 2.9879 0.6382 1.535 55.712 2.566 S14 Lens 1.6763 0.1830 2.822 S15 Filter Infinity 0.2100 2.988 S16 Infinity 0.5699 3.045 S17 Imaging Infinity 0.0084 3.298 Plane

TABLE-US-00026 TABLE 26 K A B C D E F G H S1 0.1134 -0.0041 0.0058 -0.0209 0.0313 -0.0267 0.0115 -0.0022 4E-05 S2 -31.971 -0.0343 0.0186 0.018 -0.0672 0.0895 -0.0703 0.0307 -0.0057 S3 -61.479 -0.0628 0.0273 0.0375 -0.0593 0.0197 0.0233 -0.0229 0.0061 S4 -5.9901 -0.0361 0.0057 0.1204 -0.3118 0.4572 -0.3962 0.1898 -0.0382 S5 0.4276 -0.01 -0.0036 -0.0725 0.164 -0.1983 0.1346 -0.041 0.0037 S6 -16.707 -0.0183 -0.0111 0.0191 -0.1026 0.1931 -0.1846 0.0944 -0.0194 S7 -17.469 -0.087 0.0448 -0.1275 0.1878 -0.176 0.0891 -0.0173 0.0002 S8 -45.166 -0.0908 0.0709 -0.1666 0.2236 -0.2004 0.1091 -0.0324 0.0042 S9 9.4915 -0.0528 0.0726 -0.1573 0.1702 -0.1138 0.0418 -0.0065 0 S10 27.883 -0.0964 0.0215 -0.01 0.0057 -0.0007 -0.0003 7E-05 0 S11 -18.4 0.0175 -0.105 0.0871 -0.0593 0.0265 -0.0067 0.0007 0 S12 -27.836 0.0419 -0.0472 0.0168 -0.0037 0.0005 -4E-05 1E-06 0 S13 -0.521 -0.2204 0.0759 -0.0132 0.0012 -4E-05 -3E-06 3E-07 -7E-09 S14 -0.8591 -0.2015 0.0852 -0.0311 0.0079 -0.0012 0.0001 -6E-06 1E-07

Fourteenth Example

[0216] FIG. 27 is a view illustrating a fourteenth example of an optical imaging system, and FIG. 28 illustrates aberration curves of the optical imaging system of FIG. 27.

[0217] An optical imaging system **14** includes a first lens **1014**, a second lens **2014**, a third lens **3014**, a fourth lens **4014**, a fifth lens **5014**, a sixth lens **6014**, and a seventh lens **7014**.

[0218] The first lens **1014** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2014** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3014** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4014** has a positive refractive

power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5014** has a positive refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The sixth lens **6014** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6014**. The seventh lens **7014** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens **7014**.

[0219] The optical imaging system **14** further includes a stop, a filter **8014**, and an image sensor **9014**. The stop is disposed between the first lens **1014** and the second lens **2014** to adjust an amount of light incident on the image sensor **9014**. The filter **8014** is disposed between the seventh lens **7014** and the image sensor **9014** to block infrared rays. The image sensor **9014** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **27**, the stop is disposed at a distance of 0.768 mm from the object-side surface of the first lens **1014** toward the imaging plane of the optical imaging system **14**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 14 listed in Table 59 that appears later in this application.

[0220] Table 27 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **27**, and Table 28 below shows aspherical surface coefficients of the lenses of FIG. **27**.

TABLE-US-00027 TABLE 27 Index of Effective Surface Radius of Thickness/ Refrac- Abbe Aperture No. Element Curvature Distance tion Number Radius S1 First 1.9548 0.7678 1.544 56.114 1.300 S2 Lens 7.8859 0.1005 1.209 (Stop) S3 Second 4.8288 0.2366 1.661 20.353 1.213 S4 Lens 2.8662 0.4493 1.265 S5 Third 8.8655 0.4845 1.544 56.114 1.268 S6 Lens 16.6746 0.2752 1.403 S7 Fourth 10.6715 0.3698 1.544 56.114 1.456 S8 Lens 22.4472 0.2801 1.642 S9 Fifth -4.3816 0.2711 1.661 20.353 1.769 S10 Lens -4.3828 0.1050 2.019 S11 Sixth 7.9522 0.5677 1.544 56.114 2.357 S12 Lens -3.0368 0.4648 2.647 S13 Seventh -7.5079 0.3200 1.544 56.114 3.123 S14 Lens 1.7962 0.1891 3.381 S15 Filter Infinity 0.1100 3.605 S16 Infinity 0.6749 3.635 S17 Imaging Infinity -0.0163 3.930 Plane

TABLE-US-00028 TABLE 28 K A B C D E F G H J S1 -0.8127 0.0142 0.0092 -0.0157 0.0206 -0.0137 0.0037 0.0003 -0.0003 0 S2 5.6538 -0.0472 0.0448 -0.0321 0.0158 -0.0059 0.001 0.0004 -0.0002 0 S3 -10.668 -0.0824 0.0792 -0.0266 -0.0158 0.0274 -0.0153 0.0039 -0.0004 0 S4 -0.1737 -0.0508 0.0303 0.1129 -0.3063 0.4131 -0.3101 0.1243 -0.0205 0 S5 0 -0.0377 0.0156 -0.0597 0.0773 -0.0624 0.0268 -0.0045 3E-05 0 S6 0 -0.0706 0.0482 -0.0575 -0.0009 0.0419 -0.0392 0.0166 -0.0028 0 S7 46.114 -0.1374 0.0451 0.0051 -0.0298 0.0052 0.0076 -0.0027 0.0001 0 S8 99 -0.1096 -0.0451 0.1394 -0.1519 0.0948 -0.0333 0.006 -0.0004 0 S9 -99 -0.0865 0.1152 -0.1605 0.1182 -0.0466 0.0099 -0.0011 5E-05 0 S10 -0.2245 0.0593 -0.0542 0.0004 0.0119 -0.0044 0.0007 -5E-05 1E-06 0 S11 -99 0.1031 -0.1094 0.0579 -0.0216 0.005 -0.0007 4E-05 -1E-06 0 S12 -4.7232 0.1521 -0.1221 0.0592 -0.0202 0.0046 -0.0007 5E-05 -2E-06 0 S13 -1.1986 -0.0323 -0.0724 0.0507 -0.0141 0.0021 -0.0002 8E-06 -2E-07 0 S14 -1.2644 -0.1675 0.0662 -0.0204 0.0047 -0.0007 8E-05 -5E-06 2E-07 -2E-09

Fifteenth Example

[0221] FIG. **29** is a view illustrating a fifteenth example of an optical imaging system, and FIG. **30** illustrates aberration curves of the optical imaging system of FIG. **29**.

[0222] An optical imaging system **15** includes a first lens **1015**, a second lens **2015**, a third lens **3015**, a fourth lens **4015**, a fifth lens **5015**, a sixth lens **6015**, and a seventh lens **7015**.

[0223] The first lens **1015** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2015** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is

concave. The third lens **3015** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4015** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. The fifth lens **5015** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The sixth lens **6015** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6015**. The seventh lens **7015** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens **7015**.

[0224] The optical imaging system **15** further includes a stop, a filter **8015**, and an image sensor **9015**. The stop is disposed between the first lens **1015** and the second lens **2015** to adjust an amount of light incident on the image sensor **9015**. The filter **8015** is disposed between the seventh lens **7015** and the image sensor **9015** to block infrared rays. The image sensor **9015** forms an imaging plane on which an image of a subject is formed.

[0225] Table 29 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **29**, and Table 30 below shows aspherical surface coefficients of the lenses of FIG. **29**.

TABLE-US-00029

TABLE 29	Index	Effective Surface Radius of Thickness/ Refrac- Abbe
Aperture No. Element	Curvature Distance	tion Number Radius S1 First 1.7773 0.6238 1.544
56.114 1.217 S2 Lens	6.4566 0.1000 1.158 (Stop)	S3 Second 4.4103 0.2363 1.661 20.353 1.157 S4
Lens 2.6584 0.4138 1.184 S5 Third	6.5879 0.4640 1.544 56.114 1.177 S6 Lens	10.5233 0.1777 1.282 S7 Fourth 13.4749 0.3627 1.544 56.114 1.306 S8 Lens
-20.2300 0.2325 1.444 S9 Fifth	-3.1831 0.2000 1.661 20.353 1.456 S10 Lens	-4.2151 0.1000 1.625 S11 Sixth 6.7646 0.6089 1.544 56.114 2.207
S12 Lens -2.8792 0.4211 2.145 S13 Seventh	-6.9958 0.3200 1.544 56.114 2.280 S14 Lens	1.6934 0.1485 3.165 S15 Filter Infinity 0.1100 2.850 S16 Infinity 0.7007 2.888
S17 Imaging Infinity	-0.0200 3.276 Plane	

TABLE-US-00030		TABLE 30																K	A	B	C	D	E	F	G	H	J	S1	-0.5383	0.0108	0.0209	-0.0477	0.0729																																																																																																																		
-0.06	0.0243	-0.0027	-0.0007	0	S2	5.8135	-0.0459	0.0189	0.0248	-0.0559	0.0486	-0.026	0.0094	-0.0019	0	S3	-10.011	-0.085	0.066	0.02	-0.0808	0.0756	-0.0332	0.0069	-0.0006	0	S4	-0.1875	-0.0544	0.0068	0.26	-0.6655	0.9329	-0.7519	0.3313	-0.061	0	S5	0	-0.0569	0.0063	-0.0275	-0.0046	0.0401	-0.0485	0.0264	-0.0053	0	S6	0	-0.0775	-0.0976	0.271	-0.5329	0.5567	-0.3323	0.1128	-0.0176	0	S7	47.015	-0.0863	-0.1024	0.2298	-0.2721	0.1091	0.0392	-0.0378	0.0065	0	S8	-99	-0.0603	-0.0348	0.057	-0.0468	0.0241	-0.007	0.001	-6E-05	0	S9	-99	-0.2672	0.6153	-0.9745	0.9138	-0.5236	0.1786	-0.0332	0.0026	0	S10	-0.0701	0.0268	-0.0377	-0.0253	0.035	-0.0133	0.0024	-0.0002	7E-06	0	S11	-97.721	0.1556	-0.2109	0.1424	-0.0678	0.02	-0.0033	0.0003	-1E-05	0	S12	-1.5998	0.2298	-0.1811	0.0905	-0.0342	0.0088	-0.0014	0.0001	-4E-06	0	S13	4.8341	-0.1142	-0.0024	0.0306	-0.013	0.0027	-0.0003	2E-05	-5E-07	0	S14	-1.0993	-0.2618	0.1449	-0.0599	0.0171	-0.0032	0.0004	-3E-05	1E-06	-2E-08

Sixteenth Example

[0226] FIG. **31** is a view illustrating a sixteenth example of an optical imaging system, and FIG. **32** illustrates aberration curves of the optical imaging system of FIG. **31**.

[0227] An optical imaging system **16** includes a first lens **1016**, a second lens **2016**, a third lens **3016**, a fourth lens **4016**, a fifth lens **5016**, a sixth lens **6016**, and a seventh lens **7016**.

[0228] The first lens **1016** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2016** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3016** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4016** has a positive refractive

power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5016** has a positive refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The sixth lens **6016** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6016**. The seventh lens **7016** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, no inflection point is formed on the object-side surface of the seventh lens **7016**, and one inflection point is formed on the image-side surface of the seventh lens **7016**.

[0229] The optical imaging system **16** further includes a stop, a filter **8016**, and an image sensor **9016**. The stop is disposed between the first lens **1016** and the second lens **2016** to adjust an amount of light incident on the image sensor **9016**. The filter **8016** is disposed between the seventh lens **7016** and the image sensor **9016** to block infrared rays. The image sensor **9016** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **31**, the stop is disposed at a distance of 0.641 mm from the object-side surface of the first lens **1016** toward the imaging plane of the optical imaging system **16**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 16 listed in Table 59 that appears later in this application.

[0230] Table 31 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **31**, and Table 32 below shows aspherical surface coefficients of the lenses of FIG. **31**.

TABLE-US-00031 TABLE 31 Index of Effective Surface Radius of Thickness/ Refrac- Abbe Aperture No. Element Curvature Distance tion Number Radius S1 First 1.7977 0.6409 1.544 56.114 1.270 S2 Lens 3.7422 0.1191 1.211 (Stop) S3 Second 3.0573 0.2200 1.661 20.353 1.190 S4 Lens 2.7951 0.3931 1.130 S5 Third 10.6215 0.4640 1.544 56.114 1.153 S6 Lens 9.0266 0.1000 1.289 S7 Fourth 7.9876 0.3621 1.544 56.114 1.328 S8 Lens 138.7678 0.2334 1.454 S9 Fifth -4.1765 0.2198 1.661 20.353 1.518 S10 Lens -4.1394 0.1000 1.656 S11 Sixth 4.6134 0.6089 1.544 56.114 2.000 S12 Lens -3.5921 0.4726 2.038 S13 Seventh -7.0016 0.3200 1.544 56.114 2.049 S14 Lens 1.6938 0.1107 2.685 S15 Filter Infinity 0.2100 2.942 S16 Infinity 0.5300 3.008 S17 Imaging Infinity 0.0200 3.292 Plane

TABLE-US-00032 TABLE 32 K A B C D E F G H J S1 -0.812 0.0136 0.0311 -0.0769 0.1226 -0.1099 0.0531 -0.0116 0.0005 0 S2 -6.6917 -0.0631 0.0174 0.0714 -0.1648 0.1763 -0.1086 0.0376 -0.0059 0 S3 -14.579 -0.0707 0.0068 0.1319 -0.2129 0.173 -0.0715 0.0127 -0.0005 0 S4 -0.188 -0.0614 -0.0138 0.3338 -0.7392 0.9251 -0.6781 0.276 -0.0477 0 S5 0 -0.0572 0.0435 -0.1733 0.2724 -0.2421 0.0931 -0.0042 -0.0038 0 S6 0 -0.1356 -0.0309 0.2183 -0.5547 0.6931 -0.486 0.1856 -0.0304 0 S7 30.023 -0.2107 0.0007 0.1568 -0.2854 0.2586 -0.1154 0.0236 -0.0019 0 S8 -99 -0.1858 -0.0192 0.2616 -0.4111 0.3392 -0.1538 0.0357 -0.0033 0 S9 -98.995 -0.2935 0.5043 -0.5157 0.2657 -0.0658 0.0056 0.0005 -8E-05 0 S10 -0.0701 -0.0775 0.2223 -0.2703 0.1529 -0.0452 0.0073 -0.0006 2E-05 0 S11 -97.878 0.1479 -0.1956 0.1288 -0.0598 0.0172 -0.0028 0.0002 -8E-06 0 S12 1.4166 0.1234 -0.1416 0.087 -0.0341 0.0088 -0.0014 0.0001 -4E-06 0 S13 9.5503 -0.2864 0.1096 0.0149 -0.0214 0.0064 -0.0009 6E-05 -2E-06 0 S14 -1.2786 -0.3076 0.1777 -0.0626 0.0143 -0.0022 0.0002 -1E-05 5E-07 -7E-09

Seventeenth Example

[0231] FIG. **33** is a view illustrating a seventeenth example of an optical imaging system, and FIG. **34** illustrates aberration curves of the optical imaging system of FIG. **33**.

[0232] An optical imaging system **17** includes a first lens **1017**, a second lens **2017**, a third lens **3017**, a fourth lens **4017**, a fifth lens **5017**, a sixth lens **6017**, and a seventh lens **7017**.

[0233] The first lens **1017** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2017** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is

concave. The third lens **3017** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4017** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5017** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6017** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6017**. The seventh lens **7017** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens **7017**, and one inflection point is formed on the image-side surface of the seventh lens **7017**.

[0234] The optical imaging system **17** further includes a stop, a filter **8017**, and an image sensor **9017**. The stop is disposed between the second lens **2017** and the third lens **3017** to adjust an amount of light incident on the image sensor **9017**. The filter **8017** is disposed between the seventh lens **7017** and the image sensor **9017** to block infrared rays. The image sensor **9017** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **33**, the stop is disposed at a distance of 0.9137 mm from the object-side surface of the first lens **1017** toward the imaging plane of the optical imaging system **17**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 17 listed in Table 59 that appears later in this application.

[0235] Table 33 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **33**, and Table 34 below shows aspherical surface coefficients of the lenses of FIG. **33**.

TABLE-US-00033 TABLE 33 Index of Effective Surface Radius of Thickness/ Refrac- Abbe Aperture No. Element Curvature Distance tion Number Radius S1 First 1.7485 0.6674 1.547 56.114 1.300 S2 Lens 7.7065 0.0200 1.229 S3 Second 1.8213 0.1562 1.660 20.400 1.115 S4 Lens 1.4314 0.4979 1.000 (Stop) S5 Third 4.2378 0.1600 1.660 20.400 0.993 S6 Lens 3.6200 0.1668 1.033 S7 Fourth 9.2634 1.0427 1.547 56.114 1.114 S8 Lens 28.8419 0.2336 1.480 S9 Fifth 13.9672 0.2922 1.650 21.494 1.522 S10 Lens 11.4259 0.0356 1.730 S11 Sixth 5.8246 0.4995 1.650 21.494 1.773 S12 Lens 6.8164 0.1833 2.172 S13 Seventh 1.8633 0.5958 1.537 55.711 3.150 S14 Lens 1.4257 0.2454 3.009 S15 Filter Infinity 0.1100 3.214 S16 Infinity 0.4585 3.253 S17 Imaging Infinity 0.0150 3.531 Plane

TABLE-US-00034 TABLE 34 K A B C D E F G H J S1 -0.0374 0.0041 0.0102 -0.0274 0.0363 -0.023 0.0071 -0.0007 0 0 S2 29.499 -0.0579 0.1753 -0.2761 0.2564 -0.1451 0.0461 -0.0064 0 0 S3 -1.6808 -0.0951 0.1875 -0.1212 -0.0118 0.0698 -0.0436 0.0084 0 0 S4 -0.1875 -0.0734 -0.0279 0.4656 -0.9504 1.0182 -0.5704 0.132 0 0 S5 1.3424 -0.1042 0.1034 -0.1017 0.1369 -0.1182 0.0626 -0.014 0 0 S6 6.3397 -0.1322 0.0914 -0.0544 0.0396 -0.0158 0.0024 -4E-05 0 0 S7 21.918 -0.0361 0.0211 -0.0154 0.0179 -0.009 0.0019 -0.0001 0 0 S8 25.736 -0.0408 0.011 -0.0055 0.0031 -0.0008 1E-04 -4E-06 0 0 S9 1.6857 -0.0998 0.1128 -0.1386 0.0983 -0.0424 0.0105 -0.0012 0 0 S10 -99 -0.0846 0.0566 -0.0552 0.0277 -0.0068 0.0008 -3E-05 0 0 S11 0 -0.0205 -0.0335 0.0064 0 0 0 0 0 S12 0 -0.0019 -0.0483 0.031 -0.0123 0.0029 -0.0004 2E-05 0 0 S13 -0.9427 -0.2417 0.0607 -0.0015 -0.0024 0.0006 -7E-05 4E-06 -8E-08 0 S14 -1.0048 -0.2102 0.0796 -0.0236 0.0052 -0.0008 7E-05 -3E-06 7E-08 0

Eighteenth Example

[0236] FIG. **35** is a view illustrating an eighteenth example of an optical imaging system, and FIG. **36** illustrates aberration curves of the optical imaging system of FIG. **35**.

[0237] An optical imaging system **18** includes a first lens **1018**, a second lens **2018**, a third lens **3018**, a fourth lens **4018**, a fifth lens **5018**, a sixth lens **6018**, and a seventh lens **7018**.

[0238] The first lens **1018** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2018** has a negative

refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3018** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4018** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. The fifth lens **5018** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. The sixth lens **6018** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6018**. The seventh lens **7018** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens **7018**, and one inflection point is formed on the image-side surface of the seventh lens **7018**.

[0239] The optical imaging system **18** further includes a stop, a filter **8018**, and an image sensor **9018**. The stop is disposed between the second lens **2018** and the third lens **3018** to adjust an amount of light incident on the image sensor **9018**. The filter **8018** is disposed between the seventh lens **7018** and the image sensor **9018** to block infrared rays. The image sensor **9018** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **35**, the stop is disposed at a distance of 0.8665 mm from the object-side surface of the first lens **1018** toward the imaging plane of the optical imaging system **18**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 18 listed in Table 59 that appears later in this application.

[0240] Table 35 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **35**, and Table 36 below shows aspherical surface coefficients of the lenses of FIG. **35**.

TABLE-US-00035 TABLE 35 Index of Effective Surface Radius of Thickness/ Refrac- Abbe Aperture No. Element Curvature Distance tion Number Radius S1 First 1.8575 0.6450 1.547 56.114 1.410 S2 Lens 7.5625 0.0250 1.373 S3 Second 2.3869 0.2400 1.660 20.400 1.275 S4 Lens 1.7513 0.4645 1.130 (Stop) S5 Third 4.0476 0.1300 1.660 20.400 1.092 S6 Lens 3.9656 0.2812 1.124 S7 Fourth 11.4640 0.6806 1.547 56.114 1.340 S8 Lens -165.9873 0.3664 1.596 S9 Fifth -83.3972 0.2354 1.650 21.494 1.696 S10 Lens 312.1902 0.1713 1.903 S11 Sixth 9.6436 0.8638 1.650 21.494 2.074 S12 Lens 6.8472 0.1129 2.622 S13 Seventh 2.1549 0.5985 1.537 55.711 3.411 S14 Lens 1.5639 0.2730 3.397 S15 Filter Infinity 0.1100 3.542 S16 Infinity 0.5510 3.577 S17 Imaging Infinity 0.0150 3.954 Plane

TABLE-US-00036 TABLE 36 K A B C D E F G H J S1 -0.0815 0.0055 0.0014 -0.0029 0.0061 -0.0051 0.002 -0.0003 0 0 S2 25.622 -0.0509 0.133 -0.2101 0.1935 -0.1078 0.0334 -0.0045 0 0 S3 -1.6604 -0.0775 0.1997 -0.3026 0.2917 -0.1691 0.0543 -0.0074 0 0 S4 0.011 -0.0397 0.0365 0.0313 -0.0936 0.1109 -0.0595 0.0137 0 0 S5 0.49 -0.0743 0.0501 -0.0466 0.0606 -0.0386 0.013 -0.0015 0 0 S6 7.0482 -0.0885 0.0384 -0.0259 0.0308 -0.0159 0.0036 -0.0003 0 0 S7 21.918 -0.0229 0.0042 -0.0033 0.0044 -0.0002 -0.0006 0.0001 0 0 S8 25.736 -0.0244 0.007 -0.0107 0.0065 -0.0016 0.0002 -7E-06 0 0 S9 1.6857 -0.0527 0.078 -0.0871 0.0455 -0.0137 0.0025 -0.0002 0 0 S10 76.281 -0.0552 0.0643 -0.0583 0.0243 -0.005 0.0005 -2E-05 0 0 S11 -52.836 0.0115 -0.0347 0.0203 -0.0087 0.0021 -0.0002 1E-05 0 0 S12 0 -0.0341 0.0013 0 0 0 0 0 0 S13 -0.9427 -0.1816 0.0377 -0.0008 -0.001 0.0002 -2E-05 9E-07 -2E-08 0 S14 -1.0048 -0.1579 0.0494 -0.0121 0.0022 -0.0003 2E-05 -8E-07 1E-08 0

Nineteenth Example

[0241] FIG. **37** is a view illustrating a nineteenth example of an optical imaging system, and FIG. **38** illustrates aberration curves of the optical imaging system of FIG. **37**.

[0242] An optical imaging system **19** includes a first lens **1019**, a second lens **2019**, a third lens **3019**, a fourth lens **4019**, a fifth lens **5019**, a sixth lens **6019**, and a seventh lens **7019**.

[0243] The first lens **1019** has a positive refractive power, and an object-side surface thereof is

convex and an image-side surface thereof is concave. The second lens **2019** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3019** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4019** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5019** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. The sixth lens **6019** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6019**. The seventh lens **7019** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens **7019**, and one inflection point is formed on the image-side surface of the seventh lens **7019**.

[0244] The optical imaging system **19** further includes a stop, a filter **8019**, and an image sensor **9019**. The stop is disposed between the second lens **2019** and the third lens **3019** to adjust an amount of light incident on the image sensor **9019**. The filter **8019** is disposed between the seventh lens **7019** and the image sensor **9019** to block infrared rays. The image sensor **9019** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **37**, the stop is disposed at a distance of 0.8997 mm from the object-side surface of the first lens **1019** toward the imaging plane of the optical imaging system **19**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 19 listed in Table 59 that appears later in this application.

[0245] Table 37 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **37**, and Table 38 below shows aspherical surface coefficients of the lenses of FIG. **37**.

TABLE-US-00037 TABLE 37 Index of Effective Surface Radius of Thickness/ Refrac- Abbe Aperture No. Element Curvature Distance tion Number Radius S1 First 1.8876 0.6427 1.547 56.114 1.410 S2 Lens 7.5583 0.0250 1.369 S3 Second 2.5872 0.2400 1.660 20.400 1.272 S4 Lens 1.9097 0.4388 1.130 (Stop) S5 Third 3.9821 0.1399 1.660 20.400 1.092 S6 Lens 3.8971 0.2690 1.132 S7 Fourth 10.2645 0.7282 1.547 56.114 1.352 S8 Lens 98.7612 0.3664 1.609 S9 Fifth -99.1146 0.2354 1.650 21.494 1.696 S10 Lens 195.9458 0.0598 1.903 S11 Sixth 8.0269 0.8638 1.650 21.494 2.017 S12 Lens 7.0025 0.1107 2.570 S13 Seventh 2.1549 0.6513 1.537 55.711 3.411 S14 Lens 1.5639 0.2946 3.295 S15 Filter Infinity 0.1100 3.521 S16 Infinity 0.5174 3.558 S17 Imaging Infinity 0.015 3.941 Plane

TABLE-US-00038 TABLE 38 K A B C D E F G H J S1 -0.0815 0.0055 0.0014 -0.0029 0.0061 -0.0051 0.002 -0.0003 0 0 S2 25.622 -0.0509 0.133 -0.2101 0.1935 -0.1078 0.0334 -0.0045 0 0 S3 -1.7152 -0.0796 0.199 -0.2926 0.2754 -0.1567 0.0498 -0.0068 0 0 S4 0.011 -0.0397 0.0365 0.0313 -0.0936 0.1109 -0.0595 0.0137 0 0 S5 0.49 -0.0743 0.0501 -0.0466 0.0606 -0.0386 0.013 -0.0015 0 0 S6 7.0482 -0.0885 0.0384 -0.0259 0.0308 -0.0159 0.0036 -0.0003 0 0 S7 21.918 -0.0229 0.0042 -0.0033 0.0044 -0.0002 -0.0006 0.0001 0 0 S8 25.736 -0.0244 0.007 -0.0107 0.0065 -0.0016 0.0002 -7E-06 0 0 S9 1.6857 -0.0527 0.078 -0.0871 0.0455 -0.0137 0.0025 -0.0002 0 0 S10 76.281 -0.0552 0.0643 -0.0583 0.0243 -0.005 0.0005 -2E-05 0 0 S11 -52.836 0.0115 -0.0347 0.0203 -0.0087 0.0021 -0.0002 1E-05 0 0 S12 0 -0.034 0.0045 -0.0014 0.0003 -2E-05 6E-07 0 0 0 S13 -0.9427 -0.1816 0.0377 -0.0008 -0.001 0.0002 -2E-05 9E-07 -2E-08 0 S14 -1.0048 -0.1579 0.0494 -0.0121 0.0022 -0.0003 2E-05 -8E-07 1E-08 0

Twentieth Example

[0246] FIG. **39** is a view illustrating a twentieth example of an optical imaging system, and FIG. **40** illustrates aberration curves of the optical imaging system of FIG. **39**.

[0247] An optical imaging system **20** includes a first lens **1020**, a second lens **2020**, a third lens

3020, a fourth lens **4020**, a fifth lens **5020**, a sixth lens **6020**, and a seventh lens **7020**.

[0248] The first lens **1020** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2020** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3020** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4020** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5020** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6020** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6020**. The seventh lens **7020** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens **7020**.

[0249] The optical imaging system **20** further includes a stop, a filter **8020**, and an image sensor **9020**. The stop is disposed between the second lens **2020** and the third lens **3020** to adjust an amount of light incident on the image sensor **9020**. The filter **8020** is disposed between the seventh lens **7020** and the image sensor **9020** to block infrared rays. The image sensor **9020** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **39**, the stop is disposed at a distance of 1.002 mm from the object-side surface of the first lens **1020** toward the imaging plane of the optical imaging system **20**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 20 listed in Table 59 that appears later in this application.

[0250] Table 39 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **39**, and Table 40 below shows aspherical surface coefficients of the lenses of FIG. **39**.

TABLE-US-00039 TABLE 39 Index of Effective Surface Radius of Thickness/ Refrac- Abbe Aperture No. Element Curvature Distance tion Number Radius S1 First 1.8047 0.5769 1.544 56.114 1.270 S2 Lens 5.0109 0.0406 1.230 S3 Second 4.8095 0.3545 1.544 56.114 1.204 S4 Lens 14.1878 0.0300 1.158 S5 (Stop) Third 3.6592 0.2000 1.661 20.350 1.087 S6 Lens 2.1487 0.4249 1.050 S7 Fourth 21.5791 0.3654 1.544 56.114 1.050 S8 Lens 9.6990 0.0619 1.187 S9 Fifth 6.2306 0.2825 1.639 21.525 1.212 S10 Lens 8.4970 0.3479 1.367 S11 Sixth 10.1847 0.5847 1.544 56.114 1.650 S12 Lens -1.5171 0.3562 1.934 S13 Seventh -2.7118 0.3000 1.544 56.114 2.303 S14 Lens 2.0636 0.1646 2.650 S15 Filter Infinity 0.2100 1.518 64.197 S16 Infinity 0.6300 S17 Imaging Infinity 0.0099 Plane

TABLE-US-00040 TABLE 40 K A B C D E F G H S1 -1.5984 0.022 0.0011 -0.0095 0.0071 -0.0076 0.0028 -0.0002 0 S2 0 -0.0267 -0.08 0.1204 -0.1085 0.0777 -0.0361 0.0074 0 S3 0 0.0185 -0.0944 0.1151 -0.0877 0.0713 -0.0433 0.0104 0 S4 93.032 -0.0833 0.3002 -0.6564 0.7873 -0.5697 0.2292 -0.0392 0 S5 -11.518 -0.2115 0.4874 -0.8074 0.9509 -0.7204 0.3239 -0.0644 0 S6 -4.4222 -0.0999 0.1985 -0.0999 -0.0975 0.2773 -0.2246 0.0743 0 S7 0 -0.0315 -0.1501 0.4497 -1.0958 1.4445 -1.0093 0.2957 0 S8 0 -0.1532 -0.084 0.3675 -0.5986 0.475 -0.1986 0.0366 0 S9 -76.367 -0.2472 -0.1038 0.5308 -0.6528 0.4225 -0.1503 0.0226 0 S10 0 -0.1927 -0.1015 0.3168 -0.3163 0.1912 -0.0703 0.0115 0 S11 0 0.0245 -0.0539 -0.0674 0.1082 -0.0625 0.0168 -0.0017 0 S12 -1.5099 0.2023 -0.1451 0.0004 0.0431 -0.0194 0.0035 -0.0002 0 S13 -6.0002 0.009 -0.1914 0.1596 -0.0593 0.0123 -0.0015 1E-04 -3E-06 S14 -0.8696 -0.1901 0.0765 -0.0229 0.0049 -0.0008 9E-05 -6E-06 2E-07

Twenty-First Example

[0251] FIG. **41** is a view illustrating a twenty-first example of an optical imaging system, and FIG. **42** illustrates aberration curves of the optical imaging system of FIG. **41**.

[0252] An optical imaging system **21** includes a first lens **1021**, a second lens **2021**, a third lens **3021**, a fourth lens **4021**, a fifth lens **5021**, a sixth lens **6021**, and a seventh lens **7021**.

[0253] The first lens **1021** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens **2021** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens **3021** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens **4021** has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens **5021** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens **6021** has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens **6021**. The seventh lens **7021** has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, one inflection point is formed on the object-side surface of the seventh lens **7021**, and two inflection points are formed on the image-side surface of the seventh lens **7021**.

[0254] The optical imaging system **21** further includes a stop, a filter **8021**, and an image sensor **9021**. The stop is disposed between the first lens **1021** and the second lens **2021** to adjust an amount of light incident on the image sensor **9021**. The filter **8021** is disposed between the seventh lens **7021** and the image sensor **9021** to block infrared rays. The image sensor **9021** forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. **41**, the stop is disposed at a distance of 0.374 mm from the object-side surface of the first lens **1021** toward the imaging plane of the optical imaging system **21**. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 21 listed in Table 59 that appears later in this application.

[0255] Table 41 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. **41**, and Table 42 below shows aspherical surface coefficients of the lenses of FIG. **41**.

TABLE-US-00041 TABLE 41 Index of Effective Surface Radius of Thickness/ Refrac- Abbe
Aperture No. Element Curvature Distance tion Number Radius S1 First 2.1873 0.3243 1.546
56.114 1.450 S2 Lens 1.8391 0.0497 1.441 S3 (Stop) Second 1.6361 0.7740 1.546 56.114 1.415 S4
Lens 30.6063 0.0300 1.354 S5 Third 7.2628 0.2100 1.678 19.236 1.270 S6 Lens 2.9652 0.4149
1.120 S7 Fourth 14.3312 0.3269 1.645 23.528 1.182 S8 Lens 12.1292 0.2502 1.337 S9 Fifth
2.1804 0.2500 1.645 23.528 1.580 S10 Lens 2.1733 0.3831 1.892 S11 Sixth 8.6678 0.6610 1.546
56.114 2.429 S12 Lens -1.9375 0.3110 2.544 S13 Seventh -7.6533 0.3650 1.546 56.114 2.916 S14
Lens 1.6261 0.2200 3.075 S15 Filter Infinity 0.1100 1.518 64.166 3.378 S16 Infinity 0.6351 3.414
S17 Imaging Infinity 0.0049 3.763 Plane

TABLE-US-00042 TABLE 42 K A B C D E F G H S1 -3.7488 0.0012 -0.0066 -0.0004 -0.0198
0.0252 -0.0132 0.0034 -0.0004 S2 -7.1577 -0.061 -0.0104 0.0163 0.0115 -0.0163 0.0063
-0.0009 0 S3 -2.6408 -0.0742 0.0698 -0.0582 0.0727 -0.0412 0.0034 0.0048 -0.0013 S4
-99 -0.0752 0.197 -0.3925 0.5174 -0.4377 0.2286 -0.0663 0.008 S5 0 -0.1076 0.2644
-0.4642 0.6109 -0.5485 0.3128 -0.0997 0.0134 S6 4.364 -0.0584 0.0882 -0.068 -0.0405
0.1629 -0.1817 0.0962 -0.0201 S7 0 -0.0603 0.0743 -0.2389 0.4197 -0.4882 0.353 -0.1472
0.0274 S8 0 -0.1174 0.165 -0.2983 0.348 -0.2864 0.1556 -0.0507 0.0077 S9 -15.429 -0.0562
0.0005 0.0397 -0.0576 0.0355 -0.0117 0.0015 3E-05 S10 -9.1654 -0.1003 0.0623 -0.0379
0.0141 -0.0032 5E-05 0.0002 -3E-05 S11 0 -0.001 -0.0216 0.0157 -0.0111 0.0043 -0.0009
8E-05 -3E-06 S12 -1.7327 0.1074 -0.0935 0.0649 -0.0289 0.0078 -0.0012 0.0001 -4E-06 S13
0.6082 -0.1509 0.0462 0.0036 -0.0043 0.001 -0.0001 6E-06 -2E-07 S14 -8.5925 -0.0951
0.041 -0.0124 0.0026 -0.0004 4E-05 -2E-06 4E-08

Twenty-Second Example

[0256] FIG. 43 is a view illustrating a twenty-second example of an optical imaging system, and FIG. 44 illustrates aberration curves of the optical imaging system of FIG. 43.

[0257] An optical imaging system 22 includes a first lens 1022, a second lens 2022, a third lens 3022, a fourth lens 4022, a fifth lens 5022, a sixth lens 6022, and a seventh lens 7022.

[0258] The first lens 1022 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens 2022 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens 3022 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens 4022 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens 5022 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens 6022 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens 6022. The seventh lens 7022 has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, one inflection point is formed on the object-side surface of the seventh lens 7022, and two inflection points are formed on the image-side surface of the seventh lens 7022.

[0259] The optical imaging system 22 further includes a stop, a filter 8022, and an image sensor 9022. The stop is disposed between the first lens 1022 and the second lens 2022 to adjust an amount of light incident on the image sensor 9022. The filter 8022 is disposed between the seventh lens 7022 and the image sensor 9022 to block infrared rays. The image sensor 9022 forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 43, the stop is disposed at a distance of 0.920 mm from the object-side surface of the first lens 1022 toward the imaging plane of the optical imaging system 22. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 22 listed in Table 59 that appears later in this application.

[0260] Table 43 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 43, and Table 44 below shows aspherical surface coefficients of the lenses of FIG. 43.

TABLE-US-00043 TABLE 43 Index of Effective Surface Radius of Thickness/ Refrac- Abbe
Aperture No. Element Curvature Distance tion Number Radius S1 First 1.9701 0.9200 1.546
56.114 1.435 S2 (Stop) Lens 6.1422 0.0762 1.327 S3 Second 5.3743 0.2000 1.677 19.238 1.311 S4
Lens 3.5472 0.3480 1.231 S5 Third 10.0771 0.3764 1.546 56.114 1.271 S6 Lens 25.5187 0.1640
1.351 S7 Fourth 5.8924 0.2000 1.667 20.377 1.359 S8 Lens 4.6147 0.2547 1.460 S9 Fifth 5.0940
0.2295 1.619 25.960 1.756 S10 Lens 4.3859 0.3402 1.654 S11 Sixth 4.9999 0.7714 1.546 56.114
2.420 S12 Lens -1.8739 0.3896 2.467 S13 Seventh -2.1172 0.3000 1.546 56.114 3.169 S14 Lens
2.8301 0.1800 3.066 S15 Filter Infinity 0.2100 1.518 3.715 S16 Infinity 0.6368 3.801 S17 Imaging
Infinity 0.0032 4.254 Plane

TABLE-US-00044 TABLE 44 K A B C D E F G H J S1 -1.1385 0.0141 0.023 -0.0501 0.0713
-0.0603 0.0298 -0.0079 0.0009 0 S2 12.673 -0.0899 0.0792 -0.0381 -0.0163 0.0343 -0.0229
0.0077 -0.0011 0 S3 9.9647 -0.1473 0.1118 0.0661 -0.2646 0.2998 -0.1775 0.0556 -0.0072 0
S4 -0.5888 -0.076 0.0676 0.0602 -0.1804 0.1698 -0.0679 0.0057 0.0025 0 S5 0 -0.0278
0.0424 -0.1578 0.2776 -0.3017 0.1871 -0.0609 0.0081 0 S6 -99 -0.0505 0.0344 -0.0587
0.0428 0.0016 -0.0357 0.0253 -0.0056 0 S7 0 -0.138 0.0096 0.0579 -0.2108 0.3235 -0.2566
0.1009 -0.0155 0 S8 0 -0.1363 0.1001 -0.1765 0.2075 -0.1546 0.071 -0.0193 0.0025 0 S9 0
-0.2113 0.2288 -0.2271 0.1631 -0.0851 0.0308 -0.0071 0.0008 0 S10 -62.082 -0.1439 0.0555
-0.0007 -0.029 0.0245 -0.009 0.0016 -0.0001 0 S11 -21.515 0.0047 -0.0144 0.0029 -0.0019
0.0006 -8E-05 1E-06 2E-07 0 S12 -3.7544 0.1035 -0.0491 0.0125 -0.0024 0.0003 -2E-05
-3E-07 9E-08 0 S13 -11.142 -0.0315 -0.0345 0.0239 -0.0062 0.0009 -7E-05 3E-06 -5E-08

0 S14 -1.2542 -0.091 0.025 -0.0054 0.0009 -0.0001 1E-05 -1E-06 6E-08 -1E-09

Twenty-Third Example

[0261] FIG. 45 is a view illustrating a twenty-third example of an optical imaging system, and FIG. 46 illustrates aberration curves of the optical imaging system of FIG. 45.

[0262] An optical imaging system 23 includes a first lens 1023, a second lens 2023, a third lens 3023, a fourth lens 4023, a fifth lens 5023, a sixth lens 6023, and a seventh lens 7023.

[0263] The first lens 1023 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens 2023 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens 3023 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens 4023 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens 5023 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens 6023 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is convex. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens 6023. The seventh lens 7023 has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, one inflection point is formed on the object-side surface of the seventh lens 7023, and two inflection points are formed on the image-side surface of the seventh lens 7023.

[0264] The optical imaging system 23 further includes a stop, a filter 8023, and an image sensor 9023. The stop is disposed between the first lens 1023 and the second lens 2023 to adjust an amount of light incident on the image sensor 9023. The filter 8023 is disposed between the seventh lens 7023 and the image sensor 9023 to block infrared rays. The image sensor 9023 forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 45, the stop is disposed at a distance of 1.060 mm from the object-side surface of the first lens 1023 toward the imaging plane of the optical imaging system 23. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 23 listed in Table 59 that appears later in this application.

[0265] Table 45 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 45, and Table 46 below shows aspherical surface coefficients of the lenses of FIG. 45.

TABLE-US-00045 TABLE 45 Index of Effective Surface Radius of Thickness/ Refrac- Abbe
Aperture No. Element Curvature Distance tion Number Radius S1 First 2.0122 1.0601 1.546
56.114 1.510 S2 (Stop) Lens 5.8868 0.0711 1.375 S3 Second 5.4922 0.2000 1.677 19.238
1.353 S4 Lens 3.7029 0.2969 1.240 S5 Third 9.9760 0.3703 1.546 56.114 1.270 S6 Lens
29.4320 0.1826 1.356 S7 Fourth 8.6841 0.2108 1.667 20.377 1.355 S8 Lens 6.0112 0.2603
1.479 S9 Fifth 5.7445 0.2195 1.619 25.960 1.750 S10 Lens 5.0434 0.3482 1.736 S11 Sixth
4.8476 0.9335 1.546 56.114 2.400 S12 Lens -1.7967 0.3268 2.438 S13 Seventh -1.9512
0.3000 1.546 56.114 2.692 S14 Lens 2.8062 0.1700 3.158 S15 Filter Infinity 0.2100 1.518
64.197 3.744 S16 Infinity 0.6441 3.826 S17 Imaging Infinity -0.0041 4.252 Plane

TABLE-US-00046 TABLE 46 K A B C D E F G H J S1 -1.1332 0.0154 0.0091 -0.0108 0.01
-0.0053 0.0015 -0.0002 -1E-05 0 S2 12.901 -0.0771 0.0315 0.036 -0.0911 0.0858 -0.0454
0.0132 -0.0016 0 S3 9.6379 -0.1255 0.0603 0.0962 -0.2261 0.222 -0.1214 0.036 -0.0045 0 S4
-0.8719 -0.061 0.0524 0.0027 0.0041 -0.069 0.0931 -0.0498 0.0101 0 S5 0 -0.0121 0.0151
-0.088 0.1573 -0.1697 0.1016 -0.0315 0.004 0 S6 -99 -0.0237 -0.0073 0.0365 -0.1182
0.1732 -0.1421 0.0603 -0.0104 0 S7 0 -0.104 -0.0751 0.2741 -0.5665 0.6635 -0.4435 0.1557
-0.0222 0 S8 0 -0.1088 0.0332 -0.0258 -0.0057 0.0249 -0.0175 0.0048 -0.0003 0 S9 0 -0.1824
0.192 -0.1693 0.1022 -0.0423 0.0117 -0.002 0.0002 0 S10 -96.971 -0.1318 0.0661 -0.0223
-0.0048 0.0087 -0.0034 0.0006 -4E-05 0 S11 -34.065 -0.0075 -0.0001 -0.0106 0.0081

-0.0035 0.0008 -1E-04 4E-06 0 S12 -2.7443 0.1131 -0.0726 0.0291 -0.0078 0.0012 -9E-05
1E-06 1E-07 0 S13 -10.221 -0.0393 -0.0198 0.0148 -0.0036 0.0005 -3E-05 1E-06 -2E-08 0
S14 -1.2844 -0.0938 0.0319 -0.0092 0.002 -0.0003 4E-05 -3E-06 1E-07 -3E-09

Twenty-Fourth Example

[0266] FIG. 47 is a view illustrating a twenty-fourth example of an optical imaging system, and FIG. 48 illustrates aberration curves of the optical imaging system of FIG. 47.

[0267] An optical imaging system 24 includes a first lens 1024, a second lens 2024, a third lens 3024, a fourth lens 4024, a fifth lens 5024, a sixth lens 6024, and a seventh lens 7024.

[0268] The first lens 1024 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens 2024 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens 3024 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens 4024 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens 5024 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens 6024 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens 6024. The seventh lens 7024 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens 7024, and one inflection point is formed on the image-side surface of the seventh lens 7024.

[0269] The optical imaging system 24 further includes a stop, a filter 8024, and an image sensor 9024. The stop is disposed between the second lens 2024 and the third lens 3024 to adjust an amount of light incident on the image sensor 9024. The filter 8024 is disposed between the seventh lens 7024 and the image sensor 9024 to block infrared rays. The image sensor 9024 forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 47, the stop is disposed at a distance of 1.082 mm from the object-side surface of the first lens 1024 toward the imaging plane of the optical imaging system 24. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 24 listed in Table 59 that appears later in this application.

[0270] Table 47 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 47, and Table 48 below shows aspherical surface coefficients of the lenses of FIG. 47.

TABLE-US-00047 TABLE 47 Index of Effective Surface Radius of Thickness/ Refrac- Abbe
Aperture No. Element Curvature Distance tion Number Radius S1 First 2.3369 0.4321 1.546
56.114 1.365 S2 Lens 2.8574 0.0250 1.352 S3 Second 2.5422 0.6000 1.546 56.114 1.326 S4 Lens
36.4170 0.0250 1.254 S5 (Stop) Third 8.1937 0.2300 1.679 19.236 1.217 S6 Lens 3.3336 0.3222
1.227 S7 Fourth 6.3427 0.5711 1.546 56.114 1.322 S8 Lens 11.2370 0.4049 1.372 S9 Fifth 18.9615
0.5067 1.546 56.114 1.590 S10 Lens 6.6837 0.0732 1.931 S11 Sixth 2.3548 0.6194 1.546 56.114
2.023 S12 Lens 2.5651 0.1492 2.456 S13 Seventh 1.4247 0.5400 1.546 56.114 2.710 S14 Lens
1.2822 0.3444 2.982 S15 Filter Infinity 0.2100 1.518 64.197 3.258 S16 Infinity 0.6347 3.334 S17
Imaging Infinity 0.0150 3.734 Plane

TABLE-US-00048 TABLE 48 K A B C D E F G H J S1 -0.9157 -0.0242 0.0483 -0.0925 0.0385
0.0577 -0.0925 0.0579 -0.0178 0.0022 S2 -12.376 0.0627 -0.1415 -0.3392 0.8991 -0.7358
0.1834 0.0755 -0.0533 0.0088 S3 -0.8319 0.031 -0.03 -0.6522 1.4923 -1.3976 0.6352 -0.1105
-0.0112 0.0048 S4 -7.367 -0.1852 1.7179 -6.8471 14.821 -19.261 15.464 -7.5184 2.0307
-0.2341 S5 12.337 -0.2536 1.7489 -6.6898 14.646 -19.491 16.071 -8.0307 2.2327 -0.2657 S6
1.1454 -0.0901 0.2168 -0.6218 1.4502 -2.2709 2.2634 -1.3948 0.4895 -0.0747 S7 -12.034
0.0424 -0.6838 2.5289 -5.5859 7.6559 -6.5535 3.3828 -0.9545 0.1124 S8 5.8592 -0.0168

-0.1532 0.4479 -0.9325 1.2364 -1.0356 0.5306 -0.1517 0.0187 S9 -43.521 0.0196 0.0447
-0.1445 0.1741 -0.1293 0.0589 -0.0164 0.0026 -0.0002 S10 -9.9703 -0.0233 -0.0527 0.0821
-0.0601 0.0246 -0.0062 0.001 -9E-05 4E-06 S11 -16.199 0.1383 -0.3024 0.3056 -0.2185
0.1017 -0.0304 0.0057 -0.0006 3E-05 S12 0.0118 -0.0979 0.0662 -0.0617 0.0337 -0.0119
0.0028 -0.0004 3E-05 -1E-06 S13 -0.8414 -0.3646 0.1533 -0.0353 0.0033 0.0004 -0.0001
2E-05 -8E-07 1E-08 S14 -1.4251 -0.2584 0.1351 -0.0538 0.0161 -0.0034 0.0005 -4E-05
2E-06 -4E-08

Twenty-Fifth Example

[0271] FIG. 49 is a view illustrating a twenty-fifth example of an optical imaging system, and FIG. 50 illustrates aberration curves of the optical imaging system of FIG. 49.

[0272] An optical imaging system 25 includes a first lens 1025, a second lens 2025, a third lens 3025, a fourth lens 4025, a fifth lens 5025, a sixth lens 6025, and a seventh lens 7025.

[0273] The first lens 1025 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens 2025 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens 3025 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens 4025 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens 5025 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens 6025 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens 6025. The seventh lens 7025 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens 7025.

[0274] The optical imaging system 25 further includes a stop, a filter 8025, and an image sensor 9025. The stop is disposed between the second lens 2025 and the third lens 3025 to adjust an amount of light incident on the image sensor 9025. The filter 8025 is disposed between the seventh lens 7025 and the image sensor 9025 to block infrared rays. The image sensor 9025 forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 49, the stop is disposed at a distance of 1.201 mm from the object-side surface of the first lens 1025 toward the imaging plane of the optical imaging system 25. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 25 listed in Table 59 that appears later in this application.

[0275] Table 49 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 49, and Table 50 below shows aspherical surface coefficients of the lenses of FIG. 49.

TABLE-US-00049 TABLE 49 Index of Effective Surface Radius of Thickness/ Refrac- Abbe
Aperture No. Element Curvature Distance tion Number Radius S1 First 2.2889 0.4895 1.546
56.114 1.564 S2 Lens 2.8751 0.1225 1.556 S3 Second 3.1931 0.5641 1.546 56.114 1.519 S4 Lens
102.3291 0.0250 1.495 S5 (Stop) Third 9.0291 0.2300 1.679 19.236 1.430 S6 Lens 4.0323 0.4394
1.411 S7 Fourth 6.6204 0.3813 1.546 56.114 1.543 S8 Lens 14.3245 0.5330 1.563 S9 Fifth 5.4175
0.4127 1.679 19.236 1.840 S10 Lens 3.5247 0.2029 2.415 S11 Sixth 2.3899 0.5978 1.546 56.114
2.201 S12 Lens 4.4770 0.3962 2.763 S13 Seventh 2.3256 0.5184 1.546 56.114 3.015 S14 Lens
1.4122 0.2273 3.288 S15 Filter Infinity 0.2100 1.518 64.197 3.711 S16 Infinity 0.6350 3.786 S17
Imaging Infinity 0.0150 4.203 Plane

TABLE-US-00050 TABLE 50 K A B C D E F G H J S1 -1 -0.0109 0.0161 -0.0521 0.0675
-0.0541 0.0251 -0.0062 0.0007 -2E-05 S2 -12.313 0.0249 -0.0812 0.0686 -0.0854 0.0923
-0.0564 0.019 -0.0034 0.0002 S3 -1.1961 -0.0151 -0.0414 0.0709 -0.1526 0.202 -0.1389

0.052 -0.0102 0.0008 S4 -7.0515 -0.0439 0.2205 -0.5763 0.8204 -0.7024 0.3734 -0.1213
0.0221 -0.0017 S5 9.4925 -0.0841 0.2664 -0.6308 0.9198 -0.8507 0.5017 -0.1833 0.0381
-0.0035 S6 1.6278 -0.0537 0.0672 -0.0789 0.0603 -0.0261 0.0045 0.001 -0.0003 -4E-05 S7
-4.8767 -0.0251 -0.0455 0.1569 -0.312 0.3626 -0.2555 0.1067 -0.024 0.0022 S8 5.8592
-0.0325 -0.0105 0.0226 -0.033 0.0214 -0.0047 -0.0015 0.001 -0.0001 S9 -43.521 -0.009
-0.005 0.0283 -0.0424 0.0317 -0.0144 0.004 -0.0006 4E-05 S10 -16.247 -0.0574 0.03
-0.0024 -0.0083 0.0056 -0.0019 0.0004 -4E-05 2E-06 S11 -12.323 0.0445 -0.0879 0.0791
-0.052 0.0213 -0.0055 0.0009 -8E-05 3E-06 S12 -0.1058 -0.0342 0.019 -0.0122 0.0033
-0.0005 6E-05 -9E-06 8E-07 -3E-08 S13 -0.7464 -0.2683 0.0838 -0.0065 -0.0032 0.0012
-0.0002 2E-05 -8E-07 1E-08 S14 -1.4016 -0.2382 0.1163 -0.0418 0.0106 -0.0018 0.0002
-1E-05 5E-07 -8E-09

Twenty-Sixth Example

[0276] FIG. 51 is a view illustrating a twenty-sixth example of an optical imaging system, and FIG. 52 illustrates aberration curves of the optical imaging system of FIG. 51.

[0277] An optical imaging system 26 includes a first lens 1026, a second lens 2026, a third lens 3026, a fourth lens 4026, a fifth lens 5026, a sixth lens 6026, and a seventh lens 7026.

[0278] The first lens 1026 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens 2026 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens 3026 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fourth lens 4026 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens 5026 has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The sixth lens 6026 has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens 6026. The seventh lens 7026 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens 7026.

[0279] The optical imaging system 26 further includes a stop, a filter 8026, and an image sensor 9026. The stop is disposed between the second lens 2026 and the third lens 3026 to adjust an amount of light incident on the image sensor 9026. The filter 8026 is disposed between the seventh lens 7026 and the image sensor 9026 to block infrared rays. The image sensor 9026 forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 51, the stop is disposed at a distance of 0.963 mm from the object-side surface of the first lens 1026 toward the imaging plane of the optical imaging system 26. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 26 listed in Table 59 that appears later in this application.

[0280] Table 51 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 51, and Table 52 below shows aspherical surface coefficients of the lenses of FIG. 51.

TABLE-US-00051 TABLE 51 Index of Effective Surface Radius of Thickness/ Refrac- Abbe											
Aperture No.	Element	Curvature	Distance	tion Number	Radius	S1	First	1.7493	0.7080	1.546	
56.114	1.280 S2 Lens	7.7627	0.0250	1.225	S3 Second	3.6883	0.2300	1.667	20.353	1.160	S4
(Stop)	Lens	2.4524	0.3551	1.033	S5 Third	39.9140	0.2300	1.667	20.353	1.053	S6 Lens
22.4233	0.0250 1.090 S7 Fourth	6.6877	0.3582	1.546	56.114	1.130	S8 Lens	17.1426	0.3932		
1.201	S9 Fifth	10.0343	0.3525	1.656	21.525	1.329	S10 Lens	6.5555	0.2520	1.664	S11 Sixth
-324.8644	0.6107 1.656 21.525 1.841 S12 Lens	12.2860	0.0342	2.288	S13 Seventh	1.9518					
0.8257	1.536 55.656 2.578 S14 Lens	1.7567	0.2187	2.963	S15 Filter	Infinity	0.2100	1.518			

64.197 3.258 S16 Infinity 0.6350 3.334 S17 Imaging Infinity 0.0150 3.729 Plane
TABLE-US-00052 TABLE 52 K A B C D E F G H J S1 -0.2398 5E-05 0.0225 -0.0553 0.0791
-0.0725 0.0408 -0.0137 0.0019 0 S2 6.0424 -0.0363 0.0343 0.0144 -0.1124 0.1667 -0.1307
0.054 -0.0092 0 S3 -1.7137 -0.0472 0.041 0.0264 -0.116 0.1895 -0.1701 0.0827 -0.0161 0 S4
-0.2358 -0.0167 -0.01 0.0564 -0.0195 -0.1069 0.2279 -0.1897 0.0625 0 S5 -0.0716 -0.0169
-0.0047 -0.1892 0.6295 -1.0256 0.9612 -0.4977 0.1127 0 S6 -1.1573 0.0199 -0.1372 0.1444
-0.0555 0.1408 -0.2746 0.2067 -0.0539 0 S7 -28.459 0.0213 -0.1017 0.0611 0.0456 0.018
-0.1503 0.1307 -0.0346 0 S8 -2.3038 -0.0386 0.0394 -0.1206 0.2443 -0.4112 0.4746 -0.3301
0.1229 -0.0182 S9 -3.3254 -0.1025 0.044 -0.1067 0.238 -0.3262 0.2409 -0.0929 0.0146 0
S10 -25.215 -0.0274 -0.1331 0.1909 -0.1562 0.0771 -0.0231 0.0041 -0.0003 0 S11 23.202
0.1679 -0.2882 0.2414 -0.1422 0.0533 -0.0119 0.0015 -8E-05 0 S12 -49.948 0.0068 -0.0175
0.0027 0.0001 -0.0001 4E-05 -6E-06 4E-07 0 S13 -1.9292 -0.2614 0.126 -0.0405 0.0094
-0.0015 0.0002 -9E-06 2E-07 0 S14 -0.8288 -0.1737 0.0652 -0.0206 0.0046 -0.0007 6E-05
-3E-06 7E-08 0

Twenty-Seventh Example

[0281] FIG. 53 is a view illustrating a twenty-seventh example of an optical imaging system, and FIG. 54 illustrates aberration curves of the optical imaging system of FIG. 53.

[0282] An optical imaging system 27 includes a first lens 1027, a second lens 2027, a third lens 3027, a fourth lens 4027, a fifth lens 5027, a sixth lens 6027, and a seventh lens 7027.

[0283] The first lens 1027 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens 2027 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens 3027 has a positive refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The fourth lens 4027 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens 5027 has a positive refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The sixth lens 6027 has a positive refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens 6027. The seventh lens 7027 has a negative refractive power, and an object-side surface thereof is concave and an image-side surface thereof is concave. In addition, no inflection point is formed on the object-side surface of the seventh lens 7027, and one inflection point is formed on the image-side surface of the seventh lens 7027.

[0284] The optical imaging system 27 further includes a stop, a filter 8027, and an image sensor 9027. The stop is disposed between the second lens 2027 and the third lens 3027 to adjust an amount of light incident on the image sensor 9027. The filter 8027 is disposed between the seventh lens 7027 and the image sensor 9027 to block infrared rays. The image sensor 9027 forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 53, the stop is disposed at a distance of 0.872 mm from the object-side surface of the first lens 1027 toward the imaging plane of the optical imaging system 27. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 27 listed in Table 59 that appears later in this application.

[0285] Table 53 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 53, and Table 54 below shows aspherical surface coefficients of the lenses of FIG. 53.

TABLE-US-00053 TABLE 53 Effective Surface Radius of Thickness/ Index of Abbe Aperture No.
Element Curvature Distance Refraction Number Radius S1 First 1.7603 0.6172 1.546 56.114 1.100
S2 Lens 14.1233 0.0250 1.040 S3 Second 5.8341 0.2300 1.667 20.353 1.011 S4 (Stop) Lens
3.1227 0.3733 0.919 S5 Third -49.9417 0.3799 1.546 56.114 0.995 S6 Lens -15.1870 0.1809
1.096 S7 Fourth 23.3680 0.3032 1.667 20.353 1.124 S8 Lens 12.2098 0.3354 1.309 S9 Fifth

-4.3948 0.4729 1.546 56.114 1.471 S10 Lens -1.5983 0.0250 1.698 S11 Sixth -6.0815 0.5447
 1.546 56.114 1.822 S12 Lens -3.0145 0.2724 2.192 S13 Seventh -6.1494 0.4224 1.546 56.114
 2.462 S14 Lens 1.6367 0.1933 2.880 S15 Filter Infinity 0.2100 1.518 64.197 3.223 S16 Infinity
 0.6445 3.300 S17 Imaging Infinity 0.0099 3.728 Plane
 TABLE-US-00054 TABLE 54 K A B C D E F G H J S1 -1.0054 0.0225 0.0222 -0.0696 0.1604
 -0.2238 0.1806 -0.0791 0.0141 0 S2 -1.5097 -0.1275 0.3975 -0.6982 0.6801 -0.322 0.0288
 0.029 -0.0076 0 S3 6.0294 -0.163 0.4504 -0.8514 1.0525 -0.8203 0.4235 -0.138 0.0213 0 S4
 -0.8846 -0.0449 0.0393 0.1574 -0.6934 1.3171 -1.3069 0.6799 -0.143 0 S5 0 -0.0513 -0.0193
 -0.016 0.0043 0.0034 -0.0155 0.0319 -0.0128 0 S6 0 -0.1089 -0.0569 0.3576 -0.9255 1.1947
 -0.8604 0.3322 -0.0547 0 S7 -7.5 -0.2139 -0.0107 0.1788 -0.1827 -0.1159 0.3046 -0.1897
 0.0405 0 S8 -43.341 -0.1402 -0.061 0.2777 -0.4123 0.3523 -0.1857 0.0564 -0.0071 0 S9
 -35.081 -0.0602 0.0736 -0.1046 0.1084 -0.0726 0.0255 -0.0041 0.0002 0 S10 -1.5734 0.1621
 -0.2197 0.1896 -0.107 0.0396 -0.0091 0.0011 -6E-05 0 S11 0.5153 0.2137 -0.3167 0.2399
 -0.1217 0.0384 -0.0069 0.0007 -3E-05 0 S12 -1.1466 0.1967 -0.2565 0.1542 -0.0532 0.0115
 -0.0015 0.0001 -4E-06 0 S13 -0.9056 -0.0077 -0.2094 0.1883 -0.0749 0.0167 -0.0022 0.0002
 -5E-06 0 S14 -1.2797 -0.2192 0.1006 -0.0338 0.0088 -0.0018 0.0003 -2E-05 1E-06 -3E-08
 Twenty-Eighth Example

[0286] FIG. 55 is a view illustrating a twenty-eighth example of an optical imaging system, and FIG. 56 illustrates aberration curves of the optical imaging system of FIG. 55.

[0287] An optical imaging system 28 includes a first lens 1028, a second lens 2028, a third lens 3028, a fourth lens 4028, a fifth lens 5028, a sixth lens 6028, and a seventh lens 7028.

[0288] The first lens 1028 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens 2028 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens 3028 has a positive refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The fourth lens 4028 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens 5028 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens 6028 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens 6028. The seventh lens 7028 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, two inflection points are formed on the object-side surface of the seventh lens 7028, and one inflection point is formed on the image-side surface of the seventh lens 7028.

[0289] The optical imaging system 28 further includes a stop, a filter 8028, and an image sensor 9028. The stop is disposed between the second lens 2028 and the third lens 3028 to adjust an amount of light incident on the image sensor 9028. The filter 8028 is disposed between the seventh lens 7028 and the image sensor 9028 to block infrared rays. The image sensor 9028 forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 55, the stop is disposed at a distance of 0.866 mm from the object-side surface of the first lens 1028 toward the imaging plane of the optical imaging system 28. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 28 listed in Table 59 that appears later in this application.

[0290] Table 55 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 55, and Table 56 below shows aspherical surface coefficients of the lenses of FIG. 55.

TABLE-US-00055 TABLE 55 Effective Surface Radius of Thickness/ Index of Abbe Aperture No. Element Curvature Distance Refraction Number Radius S1 First 1.8830 0.5872 1.546 56.114 1.050 S2 Lens 18.0733 0.0492 0.962 S3 Second 4.5995 0.2300 1.667 20.353 0.934 S4 (Stop)

Lens 2.5464 0.3929 0.837 S5 Third -21.7546 0.2745 1.546 56.114 1.100 S6 Lens -13.5144
 0.0611 1.106 S7 Fourth 25.3349 0.2655 1.546 56.114 1.200 S8 Lens 25.3360 0.3710 1.285 S9
 Fifth 9.4682 0.3930 1.656 21.525 1.500 S10 Lens 5.1029 0.3790 1.754 S11 Sixth 6.4162
 0.8885 1.546 56.114 2.041 S12 Lens 6.3521 0.0460 2.631 S13 Seventh 1.9665 0.8854 1.536
 55.656 3.050 S14 Lens 1.7699 0.3098 3.456 S15 Filter Infinity 0.2100 1.518 64.197 3.768 S16
 Infinity 0.6537 3.829 S17 Imaging Infinity -0.0037 4.129 Plane
 TABLE-US-00056 TABLE 56 K A B C D E F G H J S1 -0.1525 0.0035 0.0054 -0.0238 0.0587
 -0.0925 0.0808 -0.0376 0.0069 0 S2 -36.188 -0.0554 0.191 -0.4954 0.9092 -1.1194 0.849
 -0.3546 0.0617 0 S3 -0.1164 -0.0883 0.2264 -0.5273 0.9947 -1.274 1.0104 -0.4343 0.076 0 S4
 0.3326 -0.0462 0.097 -0.2316 0.5455 -0.848 0.7854 -0.3759 0.0708 0 S5 51.758 -0.0119
 -0.0911 0.3617 -0.9067 1.3845 -1.3014 0.6835 -0.1493 0 S6 42.164 0.0924 -0.5269 1.3558
 -2.2584 2.5093 -1.8107 0.7611 -0.139 0 S7 -4.7579 0.1336 -0.5938 1.261 -1.8115 1.7924
 -1.1666 0.4427 -0.0728 0 S8 -3.4393 0.0471 -0.1842 0.2886 -0.3575 0.3273 -0.1971 0.067
 -0.0093 0 S9 -8.5449 -0.0502 -0.0588 0.1599 -0.2027 0.1398 -0.0542 0.0105 -0.0007 0 S10
 -18.064 -0.044 -0.0734 0.1425 -0.1303 0.0691 -0.0217 0.0038 -0.0003 0 S11 -4.6497 0.0633
 -0.1193 0.0882 -0.0426 0.0135 -0.0028 0.0004 -2E-05 0 S12 -50 0.034 -0.0497 0.0246 -0.0072
 0.0013 -0.0001 7E-06 -2E-07 0 S13 -2.4291 -0.1201 0.0167 0.0022 -0.0009 0.0001 -6E-06
 1E-07 9E-10 0 S14 -1.0032 -0.1111 0.0248 -0.0032 -0.0001 0.0001 -2E-05 2E-06 -8E-08
 1E-09

Twenty-Ninth Example

[0291] FIG. 57 is a view illustrating a twenty-ninth example of an optical imaging system, and FIG. 58 illustrates aberration curves of the optical imaging system of FIG. 57.

[0292] An optical imaging system 29 includes a first lens 1029, a second lens 2029, a third lens 3029, a fourth lens 4029, a fifth lens 5029, a sixth lens 6029, and a seventh lens 7029.

[0293] The first lens 1029 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The second lens 2029 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The third lens 3029 has a positive refractive power, and an object-side surface thereof is concave and an image-side surface thereof is convex. The fourth lens 4029 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The fifth lens 5029 has a negative refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. The sixth lens 6029 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, at least one inflection point is formed on either one or both of the object-side surface and the image-side surface of the sixth lens 6029. The seventh lens 7029 has a positive refractive power, and an object-side surface thereof is convex and an image-side surface thereof is concave. In addition, one inflection point is formed on each of the object-side surface and the image-side surface of the seventh lens 7029.

[0294] The optical imaging system 29 further includes a stop, a filter 8029, and an image sensor 9029. The stop is disposed between the second lens 2029 and the third lens 3029 to adjust an amount of light incident on the image sensor 9029. The filter 8029 is disposed between the seventh lens 7029 and the image sensor 9029 to block infrared rays. The image sensor 9029 forms an imaging plane on which an image of a subject is formed. Although not illustrated in FIG. 57, the stop is disposed at a distance of 0.904 mm from the object-side surface of the first lens 1029 toward the imaging plane of the optical imaging system 29. This distance is equal to TTL-SL and can be calculated from the values of TTL and SL for Example 29 listed in Table 59 that appears later in this application.

[0295] Table 57 below shows physical properties of the lenses and other elements of the optical imaging system of FIG. 57, and Table 58 below shows aspherical coefficients of the lenses of FIG. 57.

TABLE-US-00057 TABLE 57 Effective Surface Radius of Thickness/ Index of Abbe Aperture No.
 Element Curvature Distance Refraction Number Radius S1 First 1.8987 0.6486 1.546 56.114 1.260
 S2 Lens 7.3568 0.0250 1.216 S3 Second 3.8789 0.2300 1.667 20.353 1.161 S4 (Stop) Lens 2.7620
 0.3408 1.053 S5 Third -50.1242 0.2819 1.546 56.114 1.120 S6 Lens -14.9889 0.0597 1.158 S7
 Fourth 12.0498 0.2698 1.546 56.114 1.220 S8 Lens 12.5657 0.2919 1.320 S9 Fifth 9.5926 0.3500
 1.667 20.353 1.520 S10 Lens 5.2748 0.3344 1.762 S11 Sixth 6.8735 0.8484 1.546 56.114 2.052
 S12 Lens 7.4933 0.0591 2.641 S13 Seventh 2.0337 0.8836 1.536 55.656 3.070 S14 Lens 1.8436
 0.3048 3.425 S15 Filter Infinity 0.2100 1.518 64.197 3.764 S16 Infinity 0.6441 3.825 S17 Imaging
 Infinity 0.0150 4.134 Plane

TABLE-US-00058 TABLE 58 K A B C D E F G H J S1 -0.1061 -0.0082 0.0469 -0.0925 0.0811
 -0.0129 -0.032 0.0224 -0.0047 0 S2 -36.188 -0.0502 0.1624 -0.4029 0.6931 -0.7643 0.5021
 -0.1789 0.0264 0 S3 0.0036 -0.0795 0.2057 -0.548 1.0742 -1.291 0.9097 -0.3412 0.052 0 S4
 0.4038 -0.0325 0.0884 -0.3009 0.7004 -0.9194 0.6738 -0.2424 0.0308 0 S5 51.758 0.0055
 -0.1746 0.5018 -0.9395 1.1442 -0.9144 0.4407 -0.0937 0 S6 42.164 0.0953 -0.4992 1.0397
 -1.2284 0.8169 -0.2802 0.0384 4E-06 0 S7 -4.7579 0.1185 -0.4938 0.8554 -0.8643 0.5167
 -0.185 0.0417 -0.0054 0 S8 -3.4393 0.0492 -0.194 0.3147 -0.3773 0.3249 -0.1878 0.063
 -0.0088 0 S9 -8.5449 -0.0638 0.0289 -0.0884 0.1649 -0.171 0.0983 -0.0306 0.0041 0 S10
 -18.064 -0.0543 -0.0172 0.0321 -0.0179 0.004 5E-06 -0.0001 8E-06 0 S11 -4.6497 0.0535
 -0.0909 0.0613 -0.0311 0.011 -0.0026 0.0004 -2E-05 0 S12 -50 0.0103 -0.0176 0.0057 -0.0015
 0.0003 -4E-05 2E-06 -6E-08 0 S13 -2.606 -0.1177 0.0192 -0.0004 -1E-04 -1E-05 4E-06
 -4E-07 9E-09 0 S14 -1.0102 -0.0979 0.0187 -0.0024 0.0001 2E-05 -6E-06 6E-07 -3E-08
 6E-10

[0296] Table 59 below shows an overall focal length f of the optical imaging system, an overall length TTL of the optical imaging system (a distance from the object-side surface of the first lens to the imaging plane), a distance SL from the stop to the imaging plane, an f-number (F No.) of the optical imaging system (the overall focal length f of the optical imaging system divided by the diameter of an entrance pupil of the optical imaging system, where both f and the diameter of the entrance pupil are expressed in mm), an image height (IMG HT) on the imaging plane (one-half of a diagonal length of the imaging plane), and a field of view (FOV) of the optical imaging system for each of Examples 1-29 described herein. The values of f, TTL, SL, and IMG HT are expressed in mm. The values of F No. are dimensionless values. The values of FOV are expressed in degrees.

TABLE-US-00059 TABLE 59 Example f TTL SL F No. IMG HT FOV 1 3.671 4.200 3.529 1.99
 3.26 81.90 2 4.780 5.827 4.402 1.57 4.05 79.02 3 4.256 5.190 3.931 1.58 3.68 80.22 4 3.950
 4.819 3.650 1.58 3.25 77.47 5 4.350 5.300 4.917 1.58 3.38 79.58 6 4.000 4.877 4.525 1.58 3.25
 76.86 7 4.280 5.100 4.369 1.71 3.54 77.84 8 4.320 5.285 4.589 1.68 3.26 72.96 9 4.401 5.300
 4.142 1.69 3.73 79.31 10 4.831 5.955 4.776 1.66 4.13 79.74 11 5.086 6.000 4.795 1.59 4.13 76.86
 12 4.447 5.144 4.894 2.07 3.53 75.63 13 4.657 5.652 5.452 2.12 3.26 69.07 14 4.700 5.650 4.882
 1.81 3.93 78.82 15 4.400 5.200 1.81 3.26 72.55 16 3.994 5.125 4.484 1.57 3.26 77.38 17 4.460
 5.450 4.5363 1.72 3.53 76.77 18 4.700 5.720 4.8535 1.67 3.93 78.23 19 4.620 5.700 4.8003 1.64
 3.93 79.17 20 4.020 4.940 3.938 1.58 3.23 76.00 21 4.333 5.320 4.946 1.50 3.75 80.30 22 4.589
 5.600 4.680 1.60 4.25 84.74 23 4.737 5.800 4.740 1.57 4.25 82.21 24 4.451 5.703 4.621 1.63 3.73
 78.60 25 4.825 6.000 4.799 1.54 4.20 80.78 26 4.592 5.478 4.515 1.79 3.73 76.90 27 4.302 5.240
 4.368 1.95 3.73 80.46 28 4.966 5.993 5.127 2.36 4.13 78.45 29 4.667 5.797 4.893 1.85 4.13 81.80

[0297] Table 60 below shows in mm a focal length f1 of the first lens, a focal length f2 of the second lens, a focal length f3 of the third lens, a focal length f4 of the fourth lens, a focal length f5 of the fifth lens, a focal length f6 of the sixth lens, and a focal length f7 of the seventh lens for each of Examples 1-29 described herein.

TABLE-US-00060 TABLE 60 Example f1 f2 f3 f4 f5 f6 f7 1 3.284 -8.740 13.548 75.065
 -13.184 44.153 -17.987 2 10.035 5.292 -7.613 22346862 86.584 -27.728 146.074 3 9.060
 4.692 -7.025 -4861.622 80.126 -24.191 1985.391 4 8.409 4.355 -6.520 -4512.292 74.369

17 -0.280 -0.330 0.880 2.410 1.440 — 18 -0.340 -0.360 0.910 2.820 1.580 — 19 -0.350 -0.360 0.950 2.800 1.600 — 20 -0.301 -0.528 0.849 — 0.718 — 21 0.202 0.201 0.967 — 0.535 0.904 22 0.334 0.378 0.719 — 0.402 0.845 23 0.354 0.431 1.004 — 0.511 0.913 24 0.210 0.372 0.610 0.706 0.722 — 25 0.199 0.269 0.603 — 0.797 — 26 0.270 0.286 0.889 — 1.015 — 27 0.276 0.509 — — 0.968 — 28 0.092 0.103 0.955 1.103 1.128 — 29 0.179 0.173 0.964 1.114 1.130 —

[0300] Table 63 below shows an inner diameter of each of the first through seventh spacers for each of Examples 1-29 described herein. S1d is an inner diameter of the first spacer SP1, S2d is an inner diameter of the second spacer SP2, S3d is an inner diameter of the third spacer SP3, S4d is an inner diameter of the fourth spacer SP4, S5d is an inner diameter of the fifth spacer SP5, S6d is an inner diameter of the sixth spacer SP6, and S7d is an inner diameter of the seventh spacer SP7.

TABLE-US-00063 TABLE 63 Example S1d S2d S3d S4d S5d S6d S7d 1 1.730 1.550 1.650 2.070 2.910 4.240 — 2 1.500 1.340 1.320 1.720 2.310 3.030 — 3 1.330 1.220 1.200 1.580 2.050 2.690 — 4 1.240 1.150 1.030 1.480 1.900 2.460 — 5 1.340 1.230 1.030 1.500 1.980 2.660 — 6 1.260 1.150 0.950 1.270 1.790 2.380 — 7 2.310 2.160 2.540 2.940 4.060 4.840 5.120 8 2.420 2.200 2.560 2.820 3.600 4.490 — 9 2.580 2.400 2.490 2.970 4.160 4.890 5.510 10 2.800 2.650 2.730 3.540 3.420 4.440 5.740 11 3.170 3.000 2.790 3.080 4.180 5.490 — 12 2.120 2.100 2.040 2.120 2.810 4.640 — 13 2.290 2.480 2.630 2.820 3.480 5.210 — 14 2.430 2.480 2.890 3.380 4.570 6.180 — 15 2.320 2.360 2.560 2.930 3.700 4.350 — 16 2.410 2.300 2.660 3.030 3.760 — — 17 2.304 2.000 2.314 3.060 3.548 5.304 — 18 2.630 2.220 2.360 3.320 4.080 5.810 — 19 2.620 2.210 2.380 3.360 4.010 5.700 — 20 2.420 2.230 2.070 2.410 3.080 4.230 — 21 2.880 2.630 2.290 2.930 4.380 5.510 — 22 2.660 2.490 2.720 3.150 4.380 5.810 — 23 2.660 2.470 — 3.130 3.780 5.130 — 24 2.680 2.510 2.540 3.000 3.960 5.280 — 25 3.070 2.920 2.900 3.320 4.400 5.750 5.930 26 2.390 2.090 2.240 2.650 3.620 4.780 5.080 27 2.060 1.890 2.150 2.700 3.610 4.560 4.840 28 1.890 1.840 2.330 2.730 3.730 5.430 6.030 29 2.390 2.150 2.400 2.820 3.940 5.680 6.020

[0301] Table 64 below shows in mm.sup.3 a volume of each of the first through seventh lenses for each of Examples 1-29 described herein. L1v is a volume of the first lens, L2v is a volume of the second lens, L3v is a volume of the third lens, L4v is a volume of the fourth lens, L5v is a volume of the fifth lens, L6v is a volume of the sixth lens, and L7v is a volume of the seventh lens.

TABLE-US-00064 TABLE 64 Example L1v L2v L3v L4v L5v L6v L7v 1 2.361 2.136 2.508 3.117 4.947 7.771 13.630 2 8.018 9.563 9.605 8.413 12.033 16.720 28.027 3 7.068 7.912 8.188 6.550 7.990 12.999 20.487 4 6.344 6.949 7.760 6.208 6.896 10.336 16.560 5 5.725 8.018 8.377 7.959 10.343 11.103 27.151 6 3.948 5.668 5.721 5.266 6.019 10.492 20.321 7 5.234 5.060 5.146 4.140 5.986 8.138 19.681 8 5.387 4.797 5.748 5.234 8.644 9.466 23.258 9 5.639 4.858 6.675 7.163 11.037 11.936 27.122 10 5.199 7.998 6.680 7.988 20.416 20.093 36.079 11 6.750 7.286 8.850 8.075 13.416 20.371 25.358 12 3.812 4.671 4.055 5.063 11.284 25.762 16.565 13 5.416 6.513 5.011 6.233 8.575 27.331 33.722 14 6.276 6.532 7.753 9.564 12.572 16.077 29.974 15 4.235 5.537 5.593 7.547 9.420 8.999 27.326 16 4.653 4.657 6.231 6.713 10.267 11.740 33.537 17 5.042 4.656 3.428 20.255 7.215 14.931 30.446 18 5.187 6.691 4.372 16.797 9.254 30.554 39.300 19 5.213 6.669 4.416 18.623 8.042 29.510 40.895 20 3.768 3.460 4.028 5.007 6.979 11.351 18.888 21 5.617 7.960 6.846 7.224 12.525 12.815 28.597 22 9.538 6.253 6.836 7.614 9.636 19.918 32.859 23 11.129 6.012 5.802 9.643 7.925 20.060 37.307 24 5.111 5.865 6.312 10.093 12.273 29.379 26.367 25 6.801 6.939 8.141 8.791 14.589 27.072 34.703 26 5.147 4.509 4.470 4.812 8.939 18.212 35.936 27 3.810 3.975 3.927 6.189 7.516 13.035 31.859 28 4.752 4.366 6.456 5.072 9.867 36.871 47.470 29 5.627 4.949 5.142 5.079 9.362 31.583 47.908

[0302] Table 65 below shows in mg a weight of each of the first through seventh lenses for each of Examples 1-29 described herein. L1w is a weight of the first lens, L2w is a weight of the second lens, L3w is a weight of the third lens, L4w is a weight of the fourth lens, L5w is a weight of the fifth lens, L6w is a weight of the sixth lens, and L7w is a weight of the seventh lens.

TABLE-US-00065 TABLE 65 Example L1w L2w L3w L4w L5w L6w L7w 1 2.455 2.628 2.608 3.242 6.184 9.714 13.767 2 8.339 9.945 12.007 10.516 12.514 20.900 28.307 3 7.351 8.229

10.235 8.188 8.310 16.249 20.692 4 6.598 7.227 9.700 7.760 7.172 12.921 16.725 5 5.954 8.339
10.472 9.710 12.619 11.547 28.237 6 4.106 5.894 7.151 6.424 7.343 10.912 21.134 7 5.444
6.223 5.351 4.306 7.362 8.463 20.468 8 5.602 5.901 5.978 5.443 10.632 9.844 24.188 9 5.865
5.052 8.344 8.953 11.478 14.920 27.393 10 5.407 8.318 8.351 8.307 21.233 25.116 37.522 11
7.020 7.578 11.062 10.093 13.952 25.464 26.372 12 3.964 5.746 4.217 5.266 14.106 26.792
17.227 13 5.633 8.011 5.212 6.483 10.547 28.424 34.059 14 6.527 8.034 8.063 9.947 15.463
16.720 31.173 15 4.404 6.810 5.817 7.849 11.587 9.359 28.419 16 4.839 5.728 6.480 6.982 12.629
12.210 34.879 17 5.244 5.726 4.216 21.065 9.018 18.664 30.751 18 5.394 8.230 5.377 17.469
11.568 38.192 39.693 19 5.422 8.203 5.432 19.368 10.052 36.887 41.304 20 3.919 3.598 4.954
5.207 8.724 11.805 19.643 21 5.842 8.279 8.558 9.030 15.657 13.327 29.741 22 9.919 7.816 7.110
9.365 11.756 20.715 34.174 23 11.574 7.515 6.034 11.861 9.669 20.863 38.799 24 5.315 6.100
7.891 10.497 12.764 30.554 27.422 25 7.073 7.217 10.176 9.142 18.237 28.155 36.091 26 5.352
5.546 5.497 5.005 11.173 22.765 36.295 27 3.962 4.889 4.084 7.612 7.817 13.556 33.133 28 4.942
5.370 6.714 5.275 12.334 38.345 47.945 29 5.852 6.087 5.348 5.282 11.516 32.847 48.387

[0303] Table 66 below shows in mm an overall outer diameter (including a rib) of each of the first to seventh lenses for each of Examples 1-29 described herein. L1TR is an overall outer diameter of the first lens, L2TR is an overall outer diameter of the second lens, L3TR is an overall outer diameter of the third lens, L4TR is an overall outer diameter of the fourth lens, L5TR is an overall outer diameter of the fifth lens, L6TR is an overall outer diameter of the sixth lens, and L7TR is an overall outer diameter of the seventh lens.

TABLE-US-00066 TABLE 66 Example L1TR L2TR L3TR L4TR L5TR L6TR L7TR 1 3.380
3.580 1.940 4.180 4.880 5.500 5.700 2 2.440 2.540 2.690 2.900 3.190 3.440 3.620 3 2.280 2.400
2.530 2.630 2.780 3.150 3.250 4 2.290 2.400 2.540 2.630 2.780 2.910 3.040 5 2.460 2.580 2.690
2.800 3.170 3.310 3.470 6 2.060 2.190 2.300 2.360 2.520 2.830 3.080 7 4.220 4.420 4.540 4.720
5.400 5.740 6.300 8 4.280 4.480 4.610 4.870 5.040 5.750 6.060 9 4.210 4.300 4.440 4.840 5.470
6.120 6.900 10 4.470 4.560 4.700 5.030 6.660 7.180 7.430 11 4.730 4.820 4.960 5.290 5.880 6.810
7.060 12 3.510 3.810 4.390 4.980 5.850 6.150 6.250 13 4.050 4.290 4.670 5.260 5.820 6.690 7.080
14 4.270 4.460 5.040 5.630 6.500 6.900 7.100 15 3.930 4.130 4.710 6.170 5.300 6.570 6.670 16
4.030 4.230 4.810 5.400 6.270 6.670 6.770 17 4.202 4.386 4.788 5.292 6.180 6.682 6.858 18 4.490
4.690 5.260 5.850 6.730 7.130 7.330 19 4.490 4.690 5.260 5.850 6.730 7.120 7.320 20 3.830 4.030
4.230 4.830 5.320 5.720 5.920 21 4.630 4.830 5.030 5.830 6.320 6.720 6.920 22 4.830 5.130 5.430
6.230 6.720 7.120 7.320 23 4.830 5.030 5.230 6.030 6.520 6.920 7.120 24 4.260 4.350 4.490 4.890
5.520 6.750 7.260 25 4.710 4.800 4.930 5.370 6.220 7.250 7.680 26 4.100 4.190 4.320 4.720 5.350
6.170 7.030 27 3.730 3.820 3.960 4.390 4.960 6.000 6.860 28 3.970 4.060 4.190 4.630 5.200 7.150
8.020 29 4.390 4.480 4.610 5.040 5.610 7.090 7.950

[0304] Table 67 below shows in mm a thickness of a flat portion of the rib of each of the first to seventh lenses for each of Examples 1-29 described herein. L1rt is a thickness of a flat portion of the rib of the first lens, L2rt is a thickness of a flat portion of the rib of the second lens, L3rt is a thickness of a flat portion of the rib of the third lens, L4rt is a thickness of a flat portion of the rib of the fourth lens, L5rt is a thickness of a flat portion of the rib of the fifth lens, L6rt is a thickness of a flat portion of the rib of the sixth lens, and L7rt is a thickness of a flat portion of the rib of the seventh lens.

TABLE-US-00067 TABLE 67 Example L1rt L2rt L3rt L4rt L5rt L6rt L7rt 1 0.245 0.260 0.255
0.240 0.305 0.275 0.405 2 0.600 0.580 0.560 0.470 0.340 0.400 0.470 3 0.600 0.540 0.540 0.440
0.250 0.380 0.420 4 0.540 0.500 0.520 0.420 0.210 0.390 0.400 5 0.390 0.440 0.470 0.360 0.420
0.380 0.470 6 0.460 0.440 0.460 0.410 0.310 0.380 0.460 7 0.435 0.430 0.360 0.215 0.320 0.330
0.405 8 0.440 0.400 0.390 0.360 0.380 0.470 0.690 9 0.550 0.380 0.580 0.410 0.500 0.320 0.530
10 0.490 0.530 0.520 0.370 0.620 0.520 0.690 11 0.600 0.440 0.600 0.460 0.580 0.600 0.440 12
0.482 0.395 0.316 0.328 0.422 0.885 0.409 13 0.634 0.643 0.375 0.356 0.312 0.828 0.984 14 0.508
0.554 0.444 0.473 0.410 0.438 0.522 15 0.431 0.556 0.361 0.429 0.380 0.380 0.667 16 0.431 0.457

0.361 0.364 0.380 0.334 0.729 17 0.439 0.446 0.280 0.866 0.225 0.359 0.763 18 0.431 0.493 0.255 0.735 0.252 0.675 0.886 19 0.431 0.493 0.255 0.790 0.198 0.637 0.886 20 0.390 0.330 0.300 0.260 0.425 0.550 0.534 21 0.540 0.480 0.460 0.250 0.555 0.395 0.688 22 0.570 0.400 0.310 0.220 0.355 0.570 0.625 23 0.620 0.400 0.270 0.430 0.300 0.520 0.760 24 0.500 0.410 0.510 0.540 0.520 0.970 0.550 25 0.530 0.410 0.560 0.570 0.480 0.600 0.620 26 0.460 0.400 0.390 0.260 0.430 0.540 0.830 27 0.400 0.420 0.370 0.500 0.320 0.460 0.720 28 0.470 0.410 0.450 0.410 0.470 0.930 0.700 29 0.440 0.390 0.400 0.400 0.380 0.740 0.720

[0305] Table 68 below shows, for each of Examples 1-29 described herein, dimensionless values of each of the ratio L1w/L7w in Conditional Expressions 1 and 6, the ratio S6d/f in Conditional Expressions 2 and 7, the ratio L1TR/L7TR in Conditional Expressions 3 and 8, the ratio L1234TRavg/L7TR in Conditional Expressions 4 and 9, and the ratio L12345TRavg/L7TR in Conditional Expressions 5 and 10. The dimensionless value of each of these ratios is obtained by dividing two values expressed in a same unit of measurement.

TABLE-US-00068 TABLE 68 L1w/ L1TR/ L1234TRavg/ L12345TRavg/ Example L7w S6d/f L7TR L7TR L7TR 1 0.178 1.155 0.5930 0.574 0.630 2 0.295 0.634 0.6740 0.7300 0.7602 3 0.355 0.632 0.7015 0.7569 0.7766 4 0.394 0.623 0.7533 0.8109 0.8316 5 0.211 0.611 0.7089 0.7586 0.7896 6 0.194 0.595 0.6688 0.7232 0.7422 7 0.266 1.131 0.6698 0.7103 0.7397 8 0.232 1.039 0.7063 0.7525 0.7683 9 0.214 1.111 0.6101 0.6446 0.6742 10 0.144 0.919 0.6016 0.6312 0.6843 11 0.266 1.079 0.6700 0.7011 0.7275 12 0.230 1.043 0.5616 0.6676 0.7213 13 0.165 1.119 0.5720 0.6451 0.6805 14 0.209 1.315 0.6014 0.6831 0.7296 15 0.155 0.989 0.5892 0.7099 0.7268 16 0.139 0.5953 0.6821 0.7309 17 0.171 1.189 0.6127 0.6805 0.7246 18 0.136 1.236 0.6126 0.6920 0.7372 19 0.131 1.234 0.6134 0.6930 0.7383 20 0.200 1.052 0.6470 0.7145 0.7514 21 0.196 1.272 0.6691 0.7341 0.7699 22 0.290 1.266 0.6598 0.7384 0.7743 23 0.298 1.083 0.6784 0.7416 0.7764 24 0.194 1.186 0.5868 0.6195 0.6477 25 0.196 1.192 0.6133 0.6449 0.6779 26 0.147 1.041 0.5832 0.6163 0.6452 27 0.120 1.060 0.5437 0.5794 0.6082 28 0.103 1.093 0.4950 0.5252 0.5499 29 0.121 1.217 0.5522 0.5824 0.6070

[0306] FIGS. 59 and 60 are cross-sectional views illustrating examples of an optical imaging system and a lens barrel coupled to each other.

[0307] The examples of an optical imaging system 100 described in this application may include a self-alignment structure as illustrated in FIGS. 59 and 60.

[0308] In one example illustrated in FIG. 59, the optical imaging system 100 includes a self-alignment structure in which optical axes of four consecutive lenses 1000, 2000, 3000, and 4000 are aligned with an optical axis of the optical imaging system 100 by coupling the four lenses 1000, 2000, 3000, and 4000 to one another.

[0309] The first lens 1000 disposed closest to object side of the optical imaging system 100 is disposed in contact with an inner surface of a lens barrel 200 to align the optical axis of the first lens 1000 with the optical axis of the optical imaging system 100, the second lens 2000 is coupled to the first lens 1000 to align the optical axis of the second lens 2000 with the optical axis of the optical imaging system 100, the third lens 3000 is coupled to the second lens 2000 to align the optical axis of the third lens 3000 with the optical axis of the optical imaging system 100, and the fourth lens 4000 is coupled to the third lens 3000 to align the optical axis of the fourth lens 4000 with the optical axis of the optical imaging system 100. The second lens 2000 to the fourth lens 4000 may not be disposed in contact with the inner surface of the lens barrel 200.

[0310] Although FIG. 59 illustrates that the first lens 1000 to the fourth lens 4000 are coupled to one another, the four consecutive lenses that are coupled to one another may be changed to the second lens 2000 to a fifth lens 5000, the third lens 3000 to a sixth lens 6000, or the fourth lens 4000 to a seventh lens 7000.

[0311] In another example illustrated in FIG. 60, the optical imaging system 100 includes a self-alignment structure in which optical axes of five consecutive lenses 1000, 2000, 3000, 4000, and 5000 are aligned with an optical axis of the optical imaging system 100 by coupling the five lenses

1000, 2000, 3000, 4000, and 5000 to one another.

[0312] The first lens **1000** disposed closest to an object side of the optical imaging system **100** is disposed in contact with an inner surface of the lens barrel **200** to align an optical axis of the first lens **1000** with the optical axis of the optical imaging system **100**, the second lens **2000** is coupled to the first lens **1000** to align the optical axis of the second lens **2000** with the optical axis of the optical imaging system **100**, the third lens **3000** is coupled to the second lens **2000** to align the optical axis of the third lens **3000** with the optical axis of the optical imaging system **100**, the fourth lens **4000** is coupled to the third lens **3000** to align the optical axis of the fourth lens **4000** with the optical axis of the optical imaging system **100**, and the fifth lens **5000** is coupled to the fourth lens **4000** to align the optical axis of the fifth lens **5000** with the optical axis of the optical imaging system **100**. The second lens **2000** to the fifth lens **5000** may not be disposed in contact with the inner surface of the lens barrel **200**.

[0313] Although FIG. **60** illustrates that the first lens **1000** to the fifth lens **5000** are coupled to one another, the five consecutive lenses that are coupled to one another may be changed to the second lens **2000** to a sixth lens **6000**, or the third lens **3000** to a seventh lens **7000**.

[0314] FIG. **61** is a cross-sectional view illustrating an example of a seventh lens.

[0315] FIG. **61** illustrates the overall outer diameter (L7TR) of the seventh lens, the thickness (L7rt) of the flat portion of the rib of the seventh lens, the thickness (L7edgeT) of the edge of the seventh lens, the thickness (Yc71P1) of the seventh lens at the first inflection point on the object-side surface of the seventh lens, the thickness (Yc71P2) of the seventh lens at the second inflection point on the object-side surface of the seventh lens, and the thickness (Yc72P1) of the seventh lens at the first inflection point on the image-side surface of the seventh lens. Although not illustrated in FIG. **61**, the seventh lens may also have a second inflection point on the image-side surface of the seventh lens, and a thickness of the seventh lens at this inflection point is Yc72P2 as listed in Table 62.

[0316] FIG. **62** is a cross-sectional view illustrating an example of a shape of a rib of a lens.

[0317] The examples of the optical imaging system **100** described in this application may include a structure for preventing a flare phenomenon and reflection.

[0318] For example, the ribs of the first to seventh lenses **1000, 2000, 3000, 4000, 5000, 6000, and 7000** of the optical imaging system may be partially surface-treated to make the surface of the rib rough as shown in FIG. **62**. Methods of surface treatment may include chemical etching, physical grinding, or any other surface treatment method capable of increasing a roughness of a surface.

[0319] A surface-treated area EA may be formed in an entire area from an edge of the optical portion of the lens through which light actually passes to an outer end of the rib. However, as illustrated in FIG. **62**, non-treated areas NEA including step portions E11, E21, and E22 may not be surface-treated, or may be surface-treated to have a roughness less than a roughness of the surface-treated area EA. The step portions E11, E21, and E22 are portions where the thickness of the rib abruptly changes. A width G1 of a first non-treated area NEA formed on an object-side surface of the lens and including a first step portion E11 may be different from a width G2 of a second non-treated area NEA formed on an image-side surface of the lens and including a second step portion E21 and a third step portion E22. In the example illustrated in FIG. **62**, G1 is greater than G2.

[0320] The width G1 of the first non-treated area NEA includes the first step portion E11, the second step portion E21, and the third step portion E22 when viewed in an optical axis direction, and the width G2 of the second non-treated area NEA includes the second step portion E21 and the third step portion E22 but not the first step portion E11 when viewed in the optical axis direction. A distance G4 from the outer end of the rib to the second step portion E21 is smaller than a distance G3 from the outer end of the rib to the first step portion E11. Also, a distance G5 from the outer end of the rib to the third step portion E22 is smaller than the distance G3 from the outer end of the rib to the first step portion E11.

[0321] The positions at which the non-treated areas NEA and the step portions E11, E21, and E22

are formed as described above and shown in FIG. 62 may be advantageous for measuring a concentricity of the lens.

[0322] The examples described above enable the optical imaging system to be miniaturized and aberrations to be easily corrected.

[0323] While this disclosure includes specific examples, it will be apparent after an understanding of the disclosure of this application that various changes in form and details may be made in these examples without departing from the spirit and scope of the claims and their equivalents. The examples described herein are to be considered in a descriptive sense only, and not for purposes of limitation. Descriptions of features or aspects in each example are to be considered as being applicable to similar features or aspects in other examples. Suitable results may be achieved if the described techniques are performed in a different order, and/or if components in a described system, architecture, device, or circuit are combined in a different manner, and/or replaced or supplemented by other components or their equivalents. Therefore, the scope of the disclosure is defined not by the detailed description, but by the claims and their equivalents, and all variations within the scope of the claims and their equivalents are to be construed as being included in the disclosure.

Claims

1. An optical imaging system comprising: a first lens comprising a refractive power; a second lens comprising a refractive power; a third lens comprising negative refractive power and a convex object-side surface; a fourth lens comprising a refractive power; a fifth lens comprising a refractive power; a sixth lens comprising a refractive power; and a seventh lens comprising a refractive power, wherein the first to seventh lens are sequentially disposed from an object side toward an imaging plane, wherein an absolute value of a radius of curvature of an image-side surface of the second lens is greater than an absolute value of a radius of curvature of an object-side surface of the first lens, wherein an absolute value of a radius of curvature of an object-side surface of the fourth lens is greater than an absolute value of a radius of curvature of an object-side surface of the third lens, and wherein an absolute value of a radius of curvature of an object-side surface of the third lens is greater than an absolute value of a radius of curvature of an image-side surface of the fifth lens.
2. The optical imaging system of claim 1, wherein the first lens has a convex object-side surface.
3. The optical imaging system of claim 1, wherein the second lens has a convex object-side surface.
4. The optical imaging system of claim 1, wherein the fourth lens has a convex object-side surface.
5. The optical imaging system of claim 1, wherein the fifth lens has a convex object-side surface.
6. The optical imaging system of claim 1, wherein the sixth lens has a convex object-side surface.
7. The optical imaging system of claim 1, wherein the seventh lens has a convex object-side surface.
8. An optical imaging system comprising: a first lens comprising a refractive power; a second lens comprising a refractive power; a third lens comprising a convex object-side surface; a fourth lens comprising a refractive power; a fifth lens comprising positive refractive power; a sixth lens comprising a refractive power; and a seventh lens comprising a refractive power, wherein the first to seventh lens are sequentially disposed from an object side toward an imaging plane, wherein an absolute value of a radius of curvature of an image-side surface of the second lens is greater than an absolute value of a radius of curvature of an object-side surface of the first lens, wherein an absolute value of a radius of curvature of an object-side surface of the fourth lens is greater than an absolute value of a radius of curvature of an object-side surface of the third lens, and wherein an absolute value of a radius of curvature of an object-side surface of the third lens is greater than an absolute value of a radius of curvature of an image-side surface of the fifth lens.
9. The optical imaging system of claim 8, wherein the first lens has a convex object-side surface.

10. The optical imaging system of claim 8, wherein the second lens has a convex object-side surface.

11. The optical imaging system of claim 8, wherein the fourth lens has a concave object-side surface.

12. The optical imaging system of claim 8, wherein the fifth lens has a concave object-side surface.

13. The optical imaging system of claim 8, wherein the sixth lens has a convex object-side surface.

14. The optical imaging system of claim 8, wherein the seventh lens has a convex object-side surface.
